There and Back Again: Bulk-to-Defect via Ward Identities

Jake Belton*[1] and Ziwen Kong†[2]

[1]*Department of Mathematics, King's College London, London, WC2R 2LS, United Kingdom*

[2]*Deutsches Elektronen-Synchrotron DESY, Notkestr. 85, 22607 Hamburg, Germany*

Abstract

In conformal field theory, the presence of a defect partially breaks the global symmetry, giving rise to defect operators such as the tilts. In this work, we derive integral identities that relate correlation functions involving bulk and defect operators—including tilts—to lower-point bulk-defect correlators, based on a detailed analysis of the Lie algebra of the symmetry group before and after the defect-induced symmetry breaking. As explicit examples, we illustrate these identities for the 1/2 BPS Maldacena-Wilson loop in $\mathcal{N} = 4$ SYM and for magnetic lines in the $O(N)$ model in $d = 4 - \varepsilon$ dimensions. We demonstrate that these identities provide a powerful tool both to check existing perturbative correlators and to impose nontrivial constraints on the CFT data.

*jake.belton@kcl.ac.uk
†zwn.kong@gmail.com

Contents

1 Introduction

Symmetry lies at the foundation of modern theoretical physics, serving as a guiding principle in the formulation and classification of quantum field theories. Among these, conformal field theories (CFTs) occupy a distinguished place, as they are defined and highly constrained by their underlying conformal symmetry. This symmetry extends the familiar Poincaré invariance to include scale transformations and special conformal transformations, leading to a rich algebraic structure that governs both the spectrum of local operators and the functional form of their correlation functions. In particular, the stringent constraints imposed by conformal invariance often render CFTs remarkably predictive, allowing many of their key properties to be determined from symmetry considerations alone.

Beyond the bulk theory, many nontrivial conformal defects are known to exist in CFTs. Such defects preserve a subalgebra of the bulk symmetry while breaking the remaining generators, providing tractable examples of symmetry breaking within a conformal framework. They generalize familiar notions of boundaries and interfaces, and often support localized degrees of freedom governed by lower-dimensional CFTs. The study of conformal defects has thus become a powerful tool for probing the structure of CFTs, offering insights into correlation functions, operator algebras, and the interplay between bulk and defect degrees of freedom [1–3]. The presence of both bulk and defect degrees of freedom enrich the theory through a larger set of correlation functions which complement bulk correlators and provide new ways to probe the theory. In this paper, we exploit classes of bulk-defect correlation functions to develop new and powerful constraints on both bulk and defect CFT data.

As a concrete example, consider a conformal defect $\mathcal{W}$ of dimension p in a d-dimensional CFT. We denote the coordinates transverse to the defect by $x_\perp$ and those along the defect by τ. Such a defect generally breaks translations in the transverse directions, which modifies the bulk stress-tensor Ward identity and gives rise to the displacement operator. This operator encodes the response of the defect to infinitesimal deformations of its position and plays a central role in characterizing its dynamics [3].

Analogously, if the bulk theory possesses an internal symmetry G that is broken to a subgroup H by the defect, the corresponding current Ward identity acquires a defect-localized modification:

$$\partial^\mu J_{\mu a}(x) = \delta^{d-p}(x_\perp) P_a^i \hat{t}_i(\tau)\,, \tag{1.1}$$

where P_a^i is the projector that decomposes the bulk symmetry indices a into the broken directions i on the defect, and $x = (\tau, x_\perp)$. The operators $\hat{t}_i$, referred to as tilt operators, inherit the defect's conformal dimension p, making them exactly marginal. Exactly marginal operators are special in CFTs because they generate continuous deformations along the conformal manifold, a feature most commonly encountered in supersymmetric theories [4–7]. Tilt operators, however, are more general: they naturally arise whenever a defect breaks an internal symmetry, and therefore appear in both supersymmetric and non-supersymmetric contexts [8–12].

The normalization of the tilt operators is fixed by the normalization of the current $J_{\mu a}$ and locally determines the Zamolodchikov metric on the defect operator space

$$g_{ij} = \langle \hat{t}_i(0)\hat{t}_j(\infty)\rangle = C_t \delta_{ij}\,, \tag{1.2}$$

where $\hat{t}(\infty) = \lim_{\tau\to\infty} |\tau|^{2p}\, \hat{t}(\tau)$. More generally, when a bulk operator $\mathcal{O}$ is present, the Ward identity is modified accordingly to

$$\partial^\mu J_{\mu a}(x)\mathcal{O}(x') = \delta^{d-p}_{\mathcal{D}}(x_\perp) P^i_a \hat{t}_i(\tau)\mathcal{O}(x') - \delta^d(x-x')(T_a\mathcal{O})(x)\,. \tag{1.3}$$

where T_a represents the action of the symmetry generator corresponding to the current $J_{\mu a}$ on the bulk operator $\mathcal{O}$. Since $\mathcal{O}$ transforms in a general representation of the symmetry group G, we label the corresponding representation space by indices α, and encode them through a polarization vector r^α.

In this way, one may define a unified generating functional that simultaneously captures the dynamics of both bulk and defect operators,

$$Z[r,w] = \int D\phi\, e^{-S} e^{\int d^d x\, r^\alpha(x)\mathcal{O}_\alpha(x)} \mathcal{W}[e^{\int d^p\tau\, w^i(\tau)\hat{t}_i(\tau)}]\,. \tag{1.4}$$

Besides the bulk coupling r^α introduced above, w_i takes values in $\mathfrak{h}^\perp \equiv \mathfrak{g}/\mathfrak{h}$, parametrizing the directions in the symmetry algebra that are broken by the presence of the defect. This functional provides a systematic framework for studying how the internal symmetry breaking influences the interplay between bulk and defect observables. In particular, connected correlators of tilt operators localized on the defect can be obtained by functional differentiation with respect to w,

$$\langle \hat{t}_{i_1}(\tau_1)\cdots\hat{t}_{i_n}(\tau_n)\rangle_c = \frac{\delta}{\delta w^{i_1}(\tau_1)}\cdots\frac{\delta}{\delta w^{i_n}(\tau_n)} \log Z[r,w]\Big|_{r=w=0}\,, \tag{1.5}$$

and more generally, one can evaluate correlators involving both bulk and defect operators,

$$\begin{aligned}&\langle \mathcal{O}_{\alpha_1}(x_1)\cdots\mathcal{O}_{\alpha_m}(x_m)\hat{t}_{i_1}(\tau_1)\cdots\hat{t}_{i_n}(\tau_n)\rangle_c\\ =&\frac{\delta}{\delta r^{\alpha_1}(x_1)}\cdots\frac{\delta}{\delta r^{\alpha_m}(x_m)}\frac{\delta}{\delta w^{i_1}(\tau_1)}\cdots\frac{\delta}{\delta w^{i_n}(\tau_n)}\log Z[r,w]\Big|_{r=w=0}\,.\end{aligned} \tag{1.6}$$

The rest of the paper is organized as follows. In Section 2, we derive integral identities that relate integrated bulk–defect correlation functions over multiple tilt operators to lower-point bulk–defect correlators. We also discuss potential scheme dependence and contact terms, and show how to construct identities that remain scheme-independent. In Sections 3 and 4, we illustrate the application of these integral identities in explicit examples, including 1/2 BPS Wilson loops in $\mathcal{N}=4$ super Yang–Mills and magnetic lines in the $O(N)$ model, and demonstrate how they constrain the CFT data. In Section 4, we further compute several new correlators that are incorporated into the integral identities.

Note: During the completion of this paper, independent work [13] appeared last week, with results that partially overlap with ours. There is another paper in progress [14] on which we heavily rely, whose conventions and techniques we adopt throughout this work.

2 Derivation

We consider the general situation in which an anomaly may arise under the action of a symmetry group G. Let L_g denote the left action of $g \in G$ on the bulk and defect couplings r and w, so that the partition function transforms as

$$Z[L_g(r), L_g(w)] = e^{A[g,w]} Z[r,w]\,, \tag{2.1}$$

where $A[g,w]$ is a local functional of w. This expression indicates that under the action of g, the partition function may not be strictly invariant but can acquire a multiplicative phase $e^{A[g,w]}$. Such a phase is known as a c-number anomaly, since it is a number-valued functional rather than an operator-valued one.

To study this systematically, we expand around the identity of the group by taking $g = e^{\lambda}$, with λ infinitesimal. Then the left action L_g can be linearized as

$$L_{e^\lambda}(w) = w + l(\lambda, w) + O(\lambda^2)\,, \qquad L_{e^\lambda}(r) = r + \rho(\lambda, r) + O(\lambda^2)\,. \tag{2.2}$$

while the anomaly functional admits a corresponding linear expansion

$$A[g,w] = \mathcal{A}[\lambda, w] + O(\lambda^2)\,. \tag{2.3}$$

Here, $l(\lambda, w)$ is a vector field linear in λ that generates infinitesimal transformations of w under the Lie algebra $\mathfrak{g}$. For elements of the unbroken subalgebra $\mathfrak{h}$, the transformation reduces to the familiar adjoint action,

$$l(\lambda, w) = [\lambda, w]\,. \tag{2.4}$$

For a general $\lambda \in \mathfrak{g}$, nonlinear effects appear, which can be organized as a formal power series in w,

$$l(\lambda, w) = \sum_{n=0}^{\infty} \frac{1}{n!} l_n(\lambda; w, \cdots, w)\,, \tag{2.5}$$

with the leading term $l_0(\lambda) = \lambda^\perp \in \mathfrak{h}^\perp$ capturing the projection of λ onto the broken directions, and higher-order terms l_n ($n \geq 1$) encoding nonlinear and renormalization–scheme–dependent contributions. Importantly, $l(\lambda, w)$ must respect the algebra $\mathfrak{g}$, which is expressed via the commutator relation

$$\left[l^i(\lambda_1, w)\frac{\partial}{\partial w^i}, l^j(\lambda_2, w)\frac{\partial}{\partial w^j} \right] = -l^i([\lambda_1, \lambda_2], w)\frac{\partial}{\partial w^i}\,. \tag{2.6}$$

Focusing on the broken directions, $\lambda \in \mathfrak{h}^\perp$, the commutator can be expanded order by order in w. Setting $\lambda_1 = \lambda$, $\lambda_2 = w$, one finds a recursive structure among the l_n,

$$\begin{aligned} & l_{n+1}(\lambda; w, w, \cdots, w) - l_{n+1}(w; \lambda, w, \cdots, w) \\ &= -\sum_{k=1}^{n} \binom{n}{k} \left(l_{n-k+1}(\lambda; l_k(w; w, \cdots, w), w, \cdots, w) - l_{n-k+1}(w; l_k(\lambda; w, \cdots, w), w, \cdots, w) \right) \\ & \quad + l_n([\lambda, w]; w, \cdots, w) \,. \end{aligned} \tag{2.7}$$

The first two explicit orders read:

$$\begin{aligned} l_1(\lambda; w) - l_1(w; \lambda) =& l_0([\lambda, w]) = [\lambda, w]^\perp \,, \\ l_2(\lambda; w, w) - l_2(w; \lambda, w) =& l_1([\lambda, w]; w) - l_1(\lambda; l_1(w; w)) + l_1(w; l_1(\lambda; w)) \,. \end{aligned} \tag{2.8}$$

The anomaly functional $\mathcal{A}[\lambda, w]$ is further constrained by the Wess–Zumino consistency condition, which ensures that the anomalous variation is compatible with the algebra $\mathfrak{g}$,

$$\int d^p\tau l^i(\lambda_1, w) \frac{\partial}{\partial w^i} \mathcal{A}[\lambda_2, w] - \int d^p\tau l^i(\lambda_2, w) \frac{\partial}{\partial w^i} \mathcal{A}[\lambda_1, w] = -\mathcal{A}[[\lambda_1, \lambda_2], w] \,. \tag{2.9}$$

Expanding $\mathcal{A}$ in powers of w,

$$\mathcal{A}[\lambda, w] = \sum_{n=0}^{\infty} \frac{1}{n!} \mathcal{A}_n[\lambda; w, \cdots, w] \,, \tag{2.10}$$

one can organize the consistency condition order by order:

$$\begin{aligned} & \sum_{k=0}^{n} \binom{n}{k} \left(\mathcal{A}_{n-k+1}[\lambda_2; l_k(\lambda_1; w, \cdots, w), w, \cdots, w] - \mathcal{A}_{n-k+1}[\lambda_1; l_k(\lambda_2; w, \cdots, w), w, \cdots, w] \right) \\ & = -\mathcal{A}_n[[\lambda_1, \lambda_2]; w, \cdots, w] \,, \end{aligned} \tag{2.11}$$

with $\mathcal{A}_0 = \mathcal{A}_1 = 0$, as discussed in [14]. The first nontrivial constraint arises at $n = 1$,

$$\mathcal{A}_2[\lambda_2; \lambda_1, w] - \mathcal{A}_2[\lambda_1; \lambda_2, w] = 0 \,. \tag{2.12}$$

Most of the conventions adopted above follow [14], where a more detailed discussion of their properties can be found. Since in this paper we are primarily interested in the bulk–defect setting, it is also necessary to specify the infinitesimal transformation of the bulk coupling r^α. This can be written as

$$\rho(\lambda, r) = \lambda^i r^\alpha \rho(e_i, e_\alpha^{(R)}) \,, \tag{2.13}$$

where $e_\alpha^{(R)}$ forms the basis of the representation R in which $\mathcal{O}$ transforms, and $\rho(e_i, e_\alpha^{(R)})$ denotes the matrix elements of the generator e_i acting on this basis, i.e., the standard representation map $T_i^{(R)}$. For convenience, in later sections we will drop the label (R).

Finally, with all these ingredients in place, we can take the logarithm of (1.4) and expanding to order r^m and w^n, one obtains identities among the connected correlators,

$$\begin{aligned}\sum_{i=1}^{m}&\langle\mathcal{O}(r_1)\cdots\mathcal{O}(\rho(\lambda,r_i))\cdots\mathcal{O}(r_m)\hat{t}(w)^n\rangle_c\\&+\sum_{k=0}^{n}\binom{n}{k}\langle\mathcal{O}(r_1)\cdots\mathcal{O}(r_m)\hat{t}(l_k(\lambda;w,\cdots,w))\hat{t}^{n-k}(w)\rangle_c=\mathcal{A}_n[\lambda;w,\cdots,w]\,.\end{aligned}\tag{2.14}$$

where we define $\mathcal{O}(r)\equiv\int d^dx\,r^\alpha(x)\mathcal{O}_\alpha(x)$, and $\hat{t}(w)\equiv\int d^p\tau\,w^i(\tau)\hat{t}_i(\tau)$. Since λ is constant, we also write $\hat{t}(\lambda)\equiv\int d^p\tau\,\lambda^i\hat{t}_i(\tau)$. From now on we focus on the case where $\lambda\in\mathfrak{h}^\perp$.

2.1 $m=0$

When $m=0$, we return to the defect-only case. A much more detailed treatment, including the anomaly terms and all associated subtleties, as well as the renormalization scheme dependence, is provided in [14]. However, to make the discussion more complete and to facilitate the use of these results in the $m\neq0$ cases later, we provide only a brief review and list the material needed subsequently.

For an n-point function, having all points on the defect leads to fewer cross-ratios compared to the general setup with m bulk and $n-m$ defect operators. This simplification provides a more tractable way to fix the l_n and $\mathcal{A}_n$. In this way, we recover the integral identities in [14]

$$\sum_{k=0}^{n}\binom{n}{k}\langle\hat{t}(l_k(\lambda;w,\cdots,w))\hat{t}^{n-k}(w)\rangle_c=\mathcal{A}_n[\lambda;w,\cdots,w]\,.\tag{2.15}$$

2.1.1 $n=0$

$$\langle\hat{t}(\lambda)\rangle_c=0\,.\tag{2.16}$$

After acting with $\frac{\delta}{\delta\lambda^i}$, the equation reads

$$\int d^p\tau_1\langle\hat{t}_i(\tau_1)\rangle_c=0\,.\tag{2.17}$$

In the case of flat defects the condition is automatically satisfied, as conformal invariance implies the vanishing of the one-point function, $\langle\hat{t}\rangle=0$. From now on, we impose this and omit defect one-point functions in the subsequent expressions.

2.1.2 $n=1$

$$\langle\hat{t}(\lambda)\hat{t}(w)\rangle_c=0\,.\tag{2.18}$$

Applying $\frac{\delta}{\delta\lambda^i}\frac{\delta}{\delta w^j(\tau_2)}$ to the above expression yields

$$\int d^p\tau_1 \langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\rangle_c = 0\,. \tag{2.19}$$

At non-coincident points, the two-point function reads

$$\langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\rangle_c = \frac{g_{ij}}{|\tau_1-\tau_2|^{2p}}\,, \tag{2.20}$$

where g_{ij} is the Zamolodchikov metric defined in (1.2), plus the distributional extension of the two-point function at coincident points $\tau_1 = \tau_2$. A more detailed analysis in [14] demonstrates that any allowed distribution is consistent with the scale invariance of the integral, ensuring that the condition is satisfied.

2.1.3 $n = 2$

$$\langle \hat{t}(\lambda)\hat{t}(w)^2\rangle_c + 2\langle \hat{t}(l_1(\lambda; w))\hat{t}(w)\rangle_c = \mathcal{A}_2[\lambda; w, w]\,. \tag{2.21}$$

At this order, the anomaly term $\mathcal{A}_2$ may be nonzero, also for the first time, the expression contains a non-vanishing, scheme-dependent term l_1. Taking the derivative $\frac{\delta}{\delta\lambda^i}\frac{\delta}{\delta w^j(\tau_2)}\frac{\delta}{\delta w^k(\tau_3)}$ of the above expression results in

$$\begin{aligned}\int d^p\tau_1 \langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c + l_1^s(e_i;e_j)\langle \hat{t}_s(\tau_2)\hat{t}_k(\tau_3)\rangle_c + l_1^s(e_i;e_k)\langle \hat{t}_s(\tau_3)\hat{t}_j(\tau_2)\rangle_c \\ = \mathcal{A}_2[e_i;e_j,e_k]\delta^p(\tau_2-\tau_3)\,.\end{aligned} \tag{2.22}$$

Generally, for arbitrary dimension p, the three-point function takes the form[1]

$$\langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle = \frac{d_{ijk}}{|\tau_1-\tau_2|^p|\tau_2-\tau_3|^p|\tau_3-\tau_1|^p}\,. \tag{2.24}$$

For any distribution at coincident points, the integral produces a non–scale-invariant contribution,

$$\langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle \ni d_{ijk}\log|\tau_2-\tau_3|\cdots\,, \tag{2.25}$$

thereby requiring $d_{ijk} = 0$. This can also be interpreted from the viewpoint of the defect conformal manifold: the three-point function of exactly marginal operators must vanish, since a nonzero value would generate a nonvanishing beta function, thereby violating the

[1] An exception is the oriented line defect with $p = 1$, where the three-point function is of the form

$$\langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle = \frac{f_{ijk}}{(\tau_1-\tau_2)(\tau_2-\tau_3)(\tau_3-\tau_1)}\,, \tag{2.23}$$

with f_{ijk} totally antisymmetric. This produces a non-vanishing anomaly term $\mathcal{A}_2$; nevertheless, as discussed in [14], the integral of such a correlator vanishes.

defining condition of conformality on the manifold. It follows that the integrated tilt three-point function vanishes identically, including at coincident points $\tau_2 = \tau_3$, and therefore $\mathcal{A}_2 = 0$ in this situation. Consequently, (2.22) transforms into

$$l_1^s(e_i; e_j)\langle \hat{t}_s(\tau_2)\hat{t}_k(\tau_3)\rangle_c + l_1^s(e_i; e_k)\langle \hat{t}_s(\tau_3)\hat{t}_j(\tau_2)\rangle_c = 0\,, \tag{2.26}$$

combined with (2.8), this allows one to solve for l_1 explicitly,

$$\begin{aligned}\langle \hat{t}_i(\tau_1)\hat{t}_s(\tau_2)\rangle l_1^s(e_j, e_k) = \frac{1}{2}\big(&\langle \hat{t}_i(\tau_1)\hat{t}_s(\tau_2)\rangle [e_j, e_k]^s + \langle \hat{t}_j(\tau_1)\hat{t}_s(\tau_2)\rangle [e_i, e_k]^s \\ &+\langle \hat{t}_k(\tau_1)\hat{t}_s(\tau_2)\rangle [e_i, e_j]^s\big)\,.\end{aligned} \tag{2.27}$$

In particular, if $[\mathfrak{h}^\perp, \mathfrak{h}^\perp] \subseteq \mathfrak{h}$—which occurs for the magnetic line in the $O(N)$ model and for 1/2 BPS Wilson loops in $\mathcal{N} = 4$ super Yang-Mills—the above expression shows that $l_1(\lambda; w) = 0$. We will use this result to simplify most of the expressions that follow.

2.1.4 $n = 3$

$$\langle \hat{t}(\lambda)\hat{t}(w)^3\rangle_c + 3\langle \hat{t}(l_1(\lambda; w))\hat{t}(w)^2\rangle_c + 3\langle \hat{t}(l_2(\lambda; w, w))\hat{t}(w)\rangle_c = \mathcal{A}_3[\lambda; w, w, w]\,. \tag{2.28}$$

We restrict to $[\mathfrak{h}^\perp, \mathfrak{h}^\perp] \subseteq \mathfrak{h}$, for which $l_1(\lambda; w) = 0$. There are, however, situations where $l_1(\lambda; w) \neq 0$; see, for example, [15, 16]. Differentiating with $\frac{\delta}{\delta\lambda^i}\frac{\delta}{\delta w^j(\tau_2)}\frac{\delta}{\delta w^k(\tau_3)}\frac{\delta}{\delta w^l(\tau_4)}$ results in

$$\begin{aligned}\int d^p\tau_1 \langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\hat{t}_m(\tau_4)\rangle_c + \delta^p(\tau_2 - \tau_3) l_2^s(e_i; e_j, e_k)\langle \hat{t}_s(\tau_2)\hat{t}_m(\tau_4)\rangle \\ +\delta^p(\tau_2 - \tau_4) l_2^s(e_i; e_j, e_m)\langle \hat{t}_s(\tau_2)\hat{t}_k(\tau_3)\rangle + \delta^p(\tau_3 - \tau_4) l_2^s(e_i; e_k, e_m)\langle \hat{t}_s(\tau_3)\hat{t}_j(\tau_2)\rangle \\ = \mathcal{A}_3[e_i; e_j, \hat{t}_k, e_m]\delta^p(\tau_2 - \tau_3)\delta^p(\tau_2 - \tau_4)\,.\end{aligned} \tag{2.29}$$

The first identity can be extracted by considering distinct insertion points. For simplicity, we set $\tau_2 = 0$, $\tau_3 = 1$, and $\tau_4 = \infty$, in which case all δ-function terms vanish and the above expression reduces to

$$\int d^p\tau_1 \langle \hat{t}_i(\tau_1)\hat{t}_j(0)\hat{t}_k(1)\hat{t}_m(\infty)\rangle_c = 0\,. \tag{2.30}$$

We should be cautious about potential divergences as τ_1 approaches τ_2, τ_3, or τ_4. Since this only-defect issue is not central to the present discussion, we refer the reader to [14] for a detailed treatment. Related considerations will, however, appear again in Section 2.2.2.

Another interesting case to consider is the double-integral setup. Let $\varphi(\tau_2)$ be an arbitrary measure satisfying $\varphi(\tau_4) = 0$. Upon integrating the above expression once more with respect to this measure, and for $\tau_3 \neq \tau_4$, we obtain

$$\int d^p\tau_1 d^p\tau_2 \varphi(\tau_2)\langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\hat{t}_m(\tau_4)\rangle_c + \varphi(\tau_3) l_2^s(e_i; e_j, e_k)\langle \hat{t}_s(\tau_3)\hat{t}_m(\tau_4)\rangle = 0\,. \tag{2.31}$$

A convenient choice is to place τ_4 at infinity, τ_3 at zero, and normalize $\varphi(\tau_3) = 1$. Although this double integration is scheme-dependent—since it corresponds to the l_2 term—it can, for the same reason, be employed to determine l_2, where

$$l_2^s(e_i; e_j, e_k) = -\int d^p\tau_1 d^p\tau_2 \varphi(\tau_2)\langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(0)\hat{t}^s(\infty)\rangle_c\,, \tag{2.32}$$

where we define $\hat{t}^s(\infty) = \frac{\delta^{rs}}{C_t}\hat{t}_r(\infty)$. In this case, we encounter not only potential divergences when two tilts coincide, but also additional divergences when three tilts collide, i.e. both $\tau_2, \tau_3 \to \tau_1$. These may lead to logarithmic divergences, necessitating regularization within a suitable renormalization scheme. The detailed discussion is deferred to Section 2.2.3, or see [14] for a more comprehensive treatment. However, by taking a commutator and using (2.8),

$$l_2(e_i; e_j, e_k) - l_2(e_j; e_i, e_k) = [[e_i, e_j], e_k]\,, \tag{2.33}$$

we find that the scheme-dependent part cancels out, leading to a scheme-independent identity

$$\int d^p\tau_1 d^p\tau_2 (\varphi(\tau_2) - \varphi(\tau_1))\langle \hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(0)\hat{t}_m(\infty)\rangle_c + [[e_i, e_j], e_k]^s\langle \hat{t}_s(0)\hat{t}_m(\infty)\rangle = 0\,. \tag{2.34}$$

Since the four-point function depends only on the cross-ratio, one of the two integrations can be performed explicitly. Consequently, in any renormalization scheme, the expression above can be reduced to a φ-independent[2] form,

$$\mathrm{vol}_{S^{p-1}} \int d^p\tau \log|\tau| \langle \hat{t}_i(1)\hat{t}_j(\tau)\hat{t}_k(0)\hat{t}_l(\infty)\rangle_c = [[e_i, e_j], e_k]^s\langle \hat{t}_s(\tau_3)\hat{t}_m(\tau_4)\rangle\,. \tag{2.35}$$

This result coincides with the integral identities relating the integrated four-tilt correlators to the curvature tensor of the defect conformal manifold, as discussed in [8, 17, 18].

2.2 $m = 1$

We next consider the situation where bulk operators are included. Note that, compared to the defect-only case 2.1, l_n now appears at order n rather than $n+1$, i.e. it is shifted down by one, due to the non-vanishing bulk-tilt functions $\langle \mathcal{O}_1(r_1)\cdots O_m(r_m)\hat{t}(l_n(\lambda; w, \cdots, w))\rangle_c$. The simplest scenario involves a single bulk operator, in which case (2.14) becomes

$$\langle \mathcal{O}(\rho(\lambda, r))\hat{t}(w)^n\rangle_c + \sum_{k=0}^{n}\binom{n}{k}\langle \mathcal{O}(r)\hat{t}(l_k(\lambda; w, \cdots, w))\hat{t}^{n-k}(w)\rangle_c = \mathcal{A}_n[\lambda; w, \cdots, w]\,. \tag{2.36}$$

[2]The result actually depends only on the value of $\varphi(0)$, which we have normalized to be 1.

2.2.1 $n = 0$

$$\langle \mathcal{O}(\rho(\lambda, r))\rangle_c + \langle \mathcal{O}(r)\hat{t}(\lambda)\rangle_c = 0\,. \tag{2.37}$$

This constitutes an identity between the bulk one-point function and the integral over the defect of the bulk–defect two-point function. Applying $\frac{\delta}{\delta\lambda^i}\frac{\delta}{\delta r^\alpha(x_0)}$ to the above, we obtain

$$(T_i)_\alpha^\beta \langle \mathcal{O}_\beta(x_0)\rangle_c + \int d^p\tau_1 \langle \mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\rangle_c = 0\,. \tag{2.38}$$

Unlike (2.17), which is rather trivial, this now provides an explicit relation between the CFT data. As an illustration, consider a bulk theory of N scalars ϕ_i $(i = 1, \cdots, N)$ with $O(N)$ symmetry, and a bulk operator $\mathcal{O}_J(u)$ of the form,

$$\mathcal{O}_J(u) = (u \cdot \phi)^J\,, \tag{2.39}$$

where u is taken to be an N-component null polarization vector, $u^2 = 0$, so that $O_J(u)$ transforms in a rank-J symmetric-traceless representation of the $O(N)$ symmetry group. This is precisely why we use u instead of r: it leaves r general, and for the same reason we use $\hat{u}$ below. Suppose there is a defect coupling to a polarization vector θ that breaks $O(N)$ to $O(N-1)$ (as occurs, for example, for 1/2 BPS Wilson loops in $\mathcal{N} = 4$ super Yang-Mills in Section 3 and for the magnetic line in the $O(N)$ model in Section 4). Then, to linear order the action of λ gives

$$\mathcal{O}_J(\rho(\lambda, u)) = -J(\lambda \cdot u)(\theta \cdot \phi)(u \cdot \phi)^{J-1}\,. \tag{2.40}$$

Recalling that the bulk one-point function and the bulk–tilt two-point function are given by

$$\begin{aligned} \langle \mathcal{O}_J(u, x_0)\rangle &= (u \cdot \theta)^J \frac{a_{\mathcal{O}_J}}{|x_{0\perp}|^{\Delta_{\mathcal{O}_J}}}\,, \\ \langle \mathcal{O}_J(u, x_0)(\hat{u} \cdot t)(\tau_1)\rangle &= (u \cdot \hat{u})(u \cdot \theta)^{J-1} \frac{b_{\mathcal{O}_J t}}{|x_{0\perp}|^{\Delta_{\mathcal{O}_J}-p}|x_{01}|^{2p}}\,. \end{aligned} \tag{2.41}$$

We choose, for convenience, $\hat{u}$ to be a null polarization vector in the preserved symmetry group $O(N-1)$, satisfying $\hat{u} \cdot \theta = \hat{u}^2 = 0$. The integral part of the bulk–tilt correlator, once integrated, takes the form

$$\int \frac{d^p\tau_1}{|x_{01}|^{2p}} = \mathrm{vol}_{S^{p-1}} \int \frac{dr r^{p-1}}{(|x_{0\perp}|^2 + r^2)^p} = \frac{2^{1-p}\pi^{\frac{p+1}{2}}}{\Gamma\left(\frac{p+1}{2}\right)|x_{0\perp}|^p}\,, \tag{2.42}$$

where Γ is Euler's gamma function, and

$$\mathrm{vol}_{S^{p-1}} = \frac{2\pi^{p/2}}{\Gamma\left(\frac{p}{2}\right)} \tag{2.43}$$

is the volume of a $(p-1)$-dimensional unit sphere S^{p-1}. Taking $\hat{u}=\lambda$ and inserting the above expressions into (2.38) yields the following constraint [19]

$$J\, a_{\mathcal{O}_J} = \frac{2^{1-p}\pi^{\frac{p+1}{2}}}{\Gamma\left(\frac{p+1}{2}\right)} b_{\mathcal{O}_J t}\,. \tag{2.44}$$

For $p=1$, which will be relevant for examples covered later, (2.44) becomes

$$J\, a_{\mathcal{O}_J} = \pi b_{\mathcal{O}_J t}\,. \tag{2.45}$$

2.2.2 $\boldsymbol{n=1}$

$$\langle \mathcal{O}(\rho(\lambda,r))\hat{t}(w)\rangle_c + \langle \mathcal{O}(r)\hat{t}(\lambda)\hat{t}(w)\rangle_c + \langle \mathcal{O}(r)\hat{t}(l_1(\lambda;w))\rangle_c = 0\,. \tag{2.46}$$

Acting with $\frac{\delta}{\delta\lambda^i}\frac{\delta}{\delta w^j(\tau_2)}\frac{\delta}{\delta r^\alpha(x_0)}$, the above expression becomes

$$(T_i)_\alpha^\beta\langle \mathcal{O}_\beta(x_0)\hat{t}_j(\tau_2)\rangle_c + \int d^p\tau_1\langle \mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\rangle_c + l_1^s(e_i;e_j)\langle \mathcal{O}_\alpha(x_0)\hat{t}_s(\tau_2)\rangle_c = 0\,. \tag{2.47}$$

In the case $[\mathfrak{h}^\perp,\mathfrak{h}^\perp]\subseteq\mathfrak{h}$ for which $l_1(e_i,e_j)=0$, this reduces to

$$(T_i)_\alpha^\beta\langle \mathcal{O}_\beta(x_0)\hat{t}_j(\tau_2)\rangle_c + \int d^p\tau_1\langle \mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\rangle_c = 0\,. \tag{2.48}$$

Returning to the bulk operator $\mathcal{O}_J$ introduced in (2.39) for $J\geq 2$, its three-point function with two tilts is given by

$$\begin{aligned}&\langle \mathcal{O}_J(u,x_0)(\hat{u}_1\cdot t)(\tau_1)(\hat{u}_2\cdot t)(\tau_2)\rangle_c\\ &= \frac{(u\cdot\theta)^{J-2}}{|x_{0\perp}|^{\Delta_{\mathcal{O}_J}-2p}|x_{01}|^{2p}|x_{02}|^{2p}}\left((u\cdot\hat{u}_1)(u\cdot\hat{u}_2)F_{1c}(\chi)+(u\cdot\theta)^2(\hat{u}_1\cdot\hat{u}_2)\frac{1}{\chi^p}F_{2c}(\chi)\right)\,,\end{aligned} \tag{2.49}$$

with one independent conformal cross-ratio

$$\chi = \frac{|x_{0\perp}|^2|\tau_{12}|^2}{|x_{01}|^2|x_{02}|^2}\,. \tag{2.50}$$

Upon contracting the indices in (2.48) with the corresponding polarization vectors u and $\hat{u}_{1,2}$, it gives

$$\begin{aligned}&\int d^p\tau_1\langle \mathcal{O}_J(u,x_0)(\hat{u}_1\cdot t)(\tau_1)(\hat{u}_2\cdot t)(\tau_2)\rangle_c\\ &= \left(J(J-1)(u\cdot\theta)^{J-2}(u\cdot\hat{u}_1)(u\cdot\hat{u}_2)-J(u\cdot\theta)^J(\hat{u}_1\cdot\hat{u}_2)\right)\frac{b_{\mathcal{O}_J t}}{|x_{0\perp}|^{\Delta_{\mathcal{O}_J}-p}|x_{02}|^{2p}}\,,\end{aligned} \tag{2.51}$$

Inserting (2.49) into the above expression and carrying out the integration, comparison of the tensor structures leads to two independent identities for F_1 and F_2,

$$\frac{1}{2}\,\mathrm{vol}_{S^{p-1}}\int_0^1 \frac{d\chi}{(\chi(1-\chi))^{1-p/2}}F_{1c}(\chi) = J(J-1)b_{\mathcal{O}_J t}\,, \tag{2.52}$$

$$\frac{1}{2}\,\mathrm{vol}_{S^{p-1}}\int_0^1 \frac{d\chi}{\chi^{1+p/2}(1-\chi)^{1-p/2}}F_{2c}(\chi) = -Jb_{\mathcal{O}_J t}\,, \tag{2.53}$$

where $\mathrm{vol}_{S^{p-1}}$ is given in (2.43).

Before proceeding, let us briefly discuss the two identities derived above. Since the correlators in (2.48) are taken to be their connected parts, whereas in much of the literature the correlators are presented in their full (including disconnected) form, we shall clarify how to extract the connected part from them. We note that for $\langle \mathcal{O}t\rangle$, the disconnected contribution vanishes, so $\langle \mathcal{O}t\rangle_c = \langle \mathcal{O}t\rangle$. In contrast, for $\langle \mathcal{O}tt\rangle$, the connected component is

$$\langle \mathcal{O}tt\rangle_c = \langle \mathcal{O}tt\rangle - \langle \mathcal{O}\rangle\langle tt\rangle\,, \tag{2.54}$$

implying that

$$F_{1c} = F_1, \qquad\qquad F_{2c} = F_2 - a_{\mathcal{O}_J}C_t\,. \tag{2.55}$$

Another subtlety arises in perturbative CFTs. If there exists an operator $\hat{\phi}$ in the θ direction appearing in the $t\times t$ OPE with scaling dimension $\Delta_{\hat{\phi}} = p+\gamma_{\hat{\phi}}$, where $\gamma_{\hat{\phi}}$ denotes the anomalous dimension at higher orders in perturbation theory, then as $\tau_1\to\tau_2$,

$$\hat{t}_i(\tau_1)\times\hat{t}_j(\tau_2) = \delta_{ij}C_t\mathbb{1}|\tau_{12}|^{-2p} + \delta_{ij}\lambda_{tt\hat{\phi}}|\tau_{12}|^{-p+\gamma_{\hat{\phi}}}\hat{\phi}(\tau_2) + O(|\tau_{12}|^0)\,. \tag{2.56}$$

Here, $\mathbb{1}$ denotes the identity operator, which is removed by subtracting the disconnected correlators. The $\hat{\phi}$ term introduces divergences upon integration over τ_1, requiring a more careful treatment of it as a distribution. Explicitly, for an arbitrary smooth test function $\varphi(\tau)$, we have

$$\int\limits_{|\tau|\le a} d^p\tau|\tau|^{-p+\gamma_{\hat{\phi}}}\varphi(|\tau|) = \mathrm{vol}_{S^{p-1}}\left(\frac{a^{\gamma_{\hat{\phi}}}}{\gamma_{\hat{\phi}}}\varphi(0) + \int\limits_0^a d\tau|\tau|^{-1+\gamma_{\hat{\phi}}}(\varphi(\tau)-\varphi(0))\right)\,, \tag{2.57}$$

For simplicity, we set $a=1$. Here we take $\varphi(\tau) = |\tau|^{-p-\gamma_{\hat{\phi}}}F_{2c}\left(\frac{\tau^2}{1+\tau^2}\right)$, which in the limit $\tau\to 0$ reduces to $b_{\mathcal{O}_J\hat{\phi}}\lambda_{tt\hat{\phi}}$. Subtracting this $\varphi(0)$ contribution renders the remaining integral on the right-hand side finite and straightforward to evaluate. Moreover, the integrand should be expanded in powers of $\gamma_{\hat{\phi}}$, leaving the integration measure as a sum of terms of the form $(\log^n\tau)/\tau$. Applying this procedure, (2.53) takes the modified form

$$\begin{aligned}\frac{1}{2}\,\mathrm{vol}_{S^{p-1}}\left(\int_0^{1/2}d\chi\left(\frac{F_{2c}(\chi)}{\chi^{1+p/2}(1-\chi)^{1-p/2}} - \frac{b_{\mathcal{O}_J\hat{\phi}}\lambda_{tt\hat{\phi}}}{\chi^{1-\gamma_{\hat{\phi}}/2}(1-\chi)^{1+\gamma_{\hat{\phi}}/2}}\right) + 2\frac{b_{\mathcal{O}_J\hat{\phi}}\lambda_{tt\hat{\phi}}}{\gamma_{\hat{\phi}}}\right)&\\ +\frac{1}{2}\,\mathrm{vol}_{S^{p-1}}\int_{1/2}^1 d\chi\frac{F_{2c}(\chi)}{\chi^{1+p/2}(1-\chi)^{1-p/2}} = -Jb_{\mathcal{O}_J t}\,,&\end{aligned} \tag{2.58}$$

where it should be noted that $\frac{1}{\chi^{1-\gamma_{\hat{\phi}}/2}(1-\chi)^{1+\gamma_{\hat{\phi}}/2}}$ must be expanded perturbatively, i.e.

$$\frac{1}{\chi^{1-\gamma_{\hat{\phi}}/2}(1-\chi)^{1+\gamma_{\hat{\phi}}/2}} = \sum_{n=0}^{\infty} \frac{\gamma_{\hat{\phi}}^n}{2^n n!} \frac{(\log\chi - \log(1-\chi))^n}{\chi(1-\chi)} \,. \tag{2.59}$$

This structure becomes particularly elegant in certain cases—for instance, for the 1/2 BPS Wilson loop in $\mathcal{N}=4$ super Yang–Mills theory—where only the contact term contributes at leading order, yielding a remarkably simple expression

$$\mathrm{vol}_{S^{p-1}} \left. \frac{b_{\mathcal{O}_J \hat{\phi}} \lambda_{tt\hat{\phi}}}{\gamma_{\hat{\phi}}} \right|_{\text{leading}} = -J b_{\mathcal{O}_J t}|_{\text{leading}} \,. \tag{2.60}$$

Furthermore, the integrated F_1 term in (2.52) at leading order also encodes simple and universal relations among CFT data. At this order, $F_1(\chi)$ reduces to a constant,

$$F_1(\chi)|_{\text{leading}} = \left(\lambda_{tt\hat{T}} b_{\mathcal{O}_J \hat{T}}\right)|_{\text{leading}} \,, \tag{2.61}$$

where $\hat{T}$ is the operator appearing in the OPE, transforming in the symmetric traceless representation,

$$\hat{t}_i(\tau_1) \times \hat{t}_j(\tau_2) \ni \frac{1}{2}\left(\delta_i^k \delta_j^l + \delta_i^l \delta_j^k - \frac{2}{N-1}\delta_{ij}\delta^{kl}\right) \lambda_{\hat{t}\hat{t}\hat{T}} |\tau_{12}|^{\gamma_{\hat{T}}} \hat{T}_{kl}(\tau_2) \,. \tag{2.62}$$

Plugging (2.61) into (2.52), we obtain another very simple expression at leading order

$$\mathrm{vol}_{S^{p-1}} \frac{\Gamma\left(\frac{p}{2}\right)^2}{2\Gamma(p)} \left(\lambda_{\hat{t}\hat{t}\hat{T}} b_{\mathcal{O}_J \hat{T}}\right)|_{\text{leading}} = J(J-1) b_{\mathcal{O}_J t}|_{\text{leading}} \,. \tag{2.63}$$

Finally, let us say a few words about the case where $[\mathfrak{h}^\perp, \mathfrak{h}^\perp] \not\subseteq \mathfrak{h}$. While l_1 itself is scheme-dependent, its commutator, as implied by (2.8), gives rise to a scheme-independent relation.

$$\begin{aligned} &\rho^\beta(e_i, e_\alpha)\langle \mathcal{O}_\beta(x_0)\hat{t}_j(\tau_2)\rangle_c - \rho^\beta(e_j, e_\alpha)\langle \mathcal{O}_\beta(x_0)\hat{t}_i(\tau_2)\rangle_c \\ &+ \left(\int d^p\tau_1 - \int d^p\tau_2\right) \langle \mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\rangle_c + [e_i, e_j]^s \langle \mathcal{O}_\alpha(x_0)\hat{t}_s(\tau_2)\rangle_c = 0 \,. \end{aligned} \tag{2.64}$$

It is straightforward to check that the terms symmetric in the indices i and j cancel, so that only the antisymmetric component remains.

2.2.3 $n = 2$

$$\begin{aligned} \langle \mathcal{O}(\rho(\lambda, r), x_0)\hat{t}(w)^2\rangle_c + \langle \mathcal{O}(r, x_0)\hat{t}(\lambda)\hat{t}^2(w)\rangle_c + 2\langle \mathcal{O}(r, x_0)\hat{t}(l_1(\lambda; w))\hat{t}(w)\rangle_c \\ + \langle \mathcal{O}(r)\hat{t}(l_2(\lambda; w, w))\rangle_c = \mathcal{A}_2[\lambda; w, w] \,. \end{aligned} \tag{2.65}$$

This is a slightly more involved case with three defect operators, where we again encounter the scheme-dependent term l_2, as in section 2.1.4. Taking the derivatives $\frac{\delta}{\delta\lambda^i}\frac{\delta}{\delta w^j(\tau_2)}\frac{\delta}{\delta w^k(\tau_3)}\frac{\delta}{\delta r^\alpha(x_0)}$, we find

$$\begin{aligned}&(T_i)^\beta_\alpha\langle\mathcal{O}_\beta(x_0)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c+\int d^p\tau_1\langle\mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c\\&+l_1^s(e_i;e_j)\langle\mathcal{O}_\alpha(x_0)\hat{t}_s(\tau_2)\hat{t}_k(\tau_3)\rangle_c+l_1^s(e_i;e_k)\langle\mathcal{O}_\alpha(x_0)\hat{t}_s(\tau_3)\hat{t}_j(\tau_2)\rangle_c\\&\qquad+\delta^p(\tau_2-\tau_3)l_2^s(e_i;e_j,e_k)\langle\mathcal{O}_\alpha(x_0)\hat{t}_s(\tau_3)\rangle_c=\delta^p(\tau_2-\tau_3)\mathcal{A}_2[e_i;e_j,e_k]\,.\end{aligned}\tag{2.66}$$

Restricting to the case $[\mathfrak{h}^\perp,\mathfrak{h}^\perp]\subseteq\mathfrak{h}$ where $l_1(e_i,e_j)=0$, similar to (2.30), we can obtain a simple identity by choosing distinct insertion points. For instance, by setting $\tau_3=0$ and $\tau_2=1$ the result simplifies to

$$(T_i)^\beta_\alpha\langle\mathcal{O}_\beta(x_0)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c+\int d^p\tau_1\langle\mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c=0\,.\tag{2.67}$$

This can be viewed as an analogue of (2.30) in the presence of a bulk operator. In order to extract the part associated with l_2 from (2.66), we perform an additional integration over τ_2 with a measure $\varphi(\tau_2)$, chosen to be the same as in (2.31), i.e., $\varphi(0)=1$ and $\varphi(\infty)=0$. Upon this integration, the above relation simplifies to

$$\begin{aligned}&(T_i)^\beta_\alpha\int d^p\tau_2\varphi(\tau_2)\langle\mathcal{O}_\beta(x_0)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c+\int d^p\tau_1d\tau_2\varphi(\tau_2)\langle\mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c\\&\qquad\qquad+\varphi(\tau_3)l_2^s(e_i;e_j,e_k)\langle\mathcal{O}_\alpha(x_0)\hat{t}_s(\tau_3)\rangle_c=\varphi(\tau_3)\mathcal{A}_2[e_i;e_j,e_k]\,.\end{aligned}\tag{2.68}$$

The integral $\int d^p\tau_1d\tau_2\varphi(\tau_2)\langle\mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\rangle_c$ involves two kinds of potentially singular regions. The first occurs when two tilt operators approach each other, which, as discussed in Section 2.2.2, requires no special regularization. The second arises when all three tilt operators collide, with the corresponding OPE given by

$$\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(\tau_3)\ni r^{-2p}\lambda^n_{ijk}(y_1,y_2)\hat{t}_n(\tau_3)+O(r^{-2p+1})\,,\tag{2.69}$$

where $y_a=(\tau_a-\tau_3)/r$ encodes the angular dependence around the coincident point, and r is the radial coordinate. The coefficients λ^n_{ijk} can be read off from the four-point functions of the tilt operators. This term may induce a logarithmic divergence, requiring regularization in this limit via an appropriate renormalization scheme, as in Section 2.1.4. Given the presence of l_2, a natural way to construct scheme-independent identities is to mimic (2.34), in which the commutators of two l_2's eliminate the scheme-dependent parts,

$$\begin{aligned}&(T_i)^\beta_\alpha\int d^p\tau_2\varphi(\tau_2)\langle\mathcal{O}_\beta(x_0)\hat{t}_j(\tau_2)\hat{t}_k(0)\rangle_c-(T_j)^\beta_\alpha\int d^p\tau_2\varphi(\tau_2)\langle\mathcal{O}_\beta(x_0)\hat{t}_i(\tau_2)\hat{t}_k(0)\rangle_c\\&+\int d^p\tau_1d\tau_2(\varphi(\tau_2)-\varphi(\tau_1))\langle\mathcal{O}_\alpha(x_0)\hat{t}_i(\tau_1)\hat{t}_j(\tau_2)\hat{t}_k(0)\rangle_c+[[e_i,e_j],e_k]^s\langle\mathcal{O}_\alpha(x_0)\hat{t}_s(0)\rangle_c=0\,,\end{aligned}\tag{2.70}$$

where the right-hand side vanishes due to (2.12), and for simplicity we set $\tau_3 = 0$. In contrast with (2.34), here the residual conformal group is too small to fully gauge away one integration variable in the double integral. Consequently, the result retains an explicit dependence on the measure $\varphi(\tau)$ and cannot be reduced to a φ-independent form like in (2.35). Nonetheless, the above expression holds for any choice of $\varphi(\tau)$ and in any renormalization scheme.

The expression for l_2 derived in the pure defect setup, (2.32), suggests an alternative route to eliminating the scheme dependence. Upon substituting (2.32), the resulting integral naturally couples pure defect correlators to those involving both bulk and defect insertions,

$$
\begin{aligned}
&(T_i)_\alpha^\beta \int d^p\tau_2 \varphi(\tau_2) \langle \mathcal{O}_\beta(x_0) \hat{t}_j(\tau_2) \hat{t}_k(0) \rangle_c \\
&+ \int d^p\tau_1 d^p\tau_2 \varphi(\tau_2) (\langle \mathcal{O}_\alpha(x_0) \hat{t}_i(\tau_1) \hat{t}_j(\tau_2) \hat{t}_k(0) \rangle_c - \langle \hat{t}_i(\tau_1) \hat{t}_j(\tau_2) \hat{t}_k(0) \hat{t}^s(\infty) \rangle_c \langle \mathcal{O}_\alpha(x_0) \hat{t}_s(0) \rangle_c) \\
&= \mathcal{A}_2[e_i; e_j, e_k] \, .
\end{aligned}
\tag{2.71}
$$

One of the advantages of this expression is that the potential logarithmic divergences (2.69) in the 1-bulk–3-tilt correlator and the tilt four-point function cancel each other exactly, so no regularization is required.

2.3 $m \geq 2$

Similar constraints can, of course, be constructed for bulk–defect correlators in (2.14) with more bulk insertions. However, these do not yield genuinely new CFT data if all bulk three-point functions are already known. Indeed, the OPE of two operators transforming in representations R_1 and R_2 of the internal symmetry group can only involve exchanged operators belonging to irreducible representations appearing in the Clebsch–Gordan decomposition of $R_1 \otimes R_2$. Since the bulk operators considered in (2.36) are completely general and can transform in any representation, one can always apply the OPE to reduce a product of bulk operators to a sum over single operators in irreducible representations. Consequently, the analysis performed in Section (2.2)—for the single bulk insertion—already captures all independent constraints, and higher-bulk identities can be viewed as recursive applications of the same logic.

In practice, however, knowing all three-point functions of all operators is possible only in a few very simple cases, such as free theories or certain two-dimensional rational CFTs [20–22]. In generic CFTs, this information is not available, so the identities in (2.14) become a powerful tool: they can be incorporated into bootstrap algorithms, both analytically and numerically, to generate genuinely new constraints on the CFT data.

In particular, the bulk–defect setup provides a rich framework for extracting CFT data. In addition to the familiar bulk-channel OPE, there exists an independent channel—the bulk–defect OPE—which describes how a bulk operator, when brought close to the defect, becomes indistinguishable from a defect operator [3]. This process is governed by the bulk-to-defect OPE coefficients $b_{\mathcal{O}\hat{\mathcal{O}}}$,

$$\mathcal{O}(x) = \sum_{\hat{\mathcal{O}}} b_{\mathcal{O}\hat{\mathcal{O}}} |x_\perp|^{\Delta_{\mathcal{O}} - \Delta_{\hat{\mathcal{O}}}} (\hat{\mathcal{O}}(\tau) + \text{descendants}) \,. \tag{2.72}$$

In the last sections, we have already discussed how the $m = 1$ relation constrains these CFT data, see (2.44) (2.58) (2.60) and (2.63). We expect that the equations with $m \geq 2$ will impose additional, independent constraints, which we leave for future investigation.

3 Example: 1/2 BPS Wilson lines in $\mathcal{N} = 4$ SYM

In this section, we illustrate the integral identities derived above using the example of the 1/2 BPS Wilson loop in $\mathcal{N} = 4$ super Yang–Mills theory [23, 24], which takes the form

$$\mathcal{W} = \frac{1}{N} \operatorname{tr} \mathcal{P} \exp \int (i A_\mu \dot{x}^\mu + |\dot{x}| \theta \cdot \phi) \, d\tau \,, \tag{3.1}$$

where we set $\theta^i = \delta^{i6}$, The remaining five scalars then play the role of the tilt operators appearing in (1.1) [25–30], normalized as

$$C_{\hat{t}} = \frac{\sqrt{\lambda}}{2\pi^2} \frac{I_2(\sqrt{\lambda})}{I_1(\sqrt{\lambda})} \,, \tag{3.2}$$

where $C_{\hat{t}}$ is defined in (1.2), and I_n are the modified Bessel functions. The bulk operators we consider are the single-trace operators, constructed similarly to (2.40), but with a constant prefactor:

$$\mathcal{O}_J(u, x) = (2\pi)^J \frac{2^{J/2}}{\sqrt{J \lambda^J}} \operatorname{tr} (u \cdot \phi(x))^J \tag{3.3}$$

Here again we consider the case $J \geq 2$. The null vector u guarantees that $\mathcal{O}_J$ transforms in a rank-J symmetric-traceless representation of the R-symmetry group. These operators are protected and have scaling dimensions equal to their R-charge J.

3.1 $\int \langle \mathcal{O} \hat{t} \rangle \sim \langle \mathcal{O} \rangle$

The first identity that we want to demonstrate in this theory is (2.45), which can be rewritten simply as

$$\frac{J}{\pi} a_{\mathcal{O}_J} = b_{\mathcal{O}_J \hat{t}} \,, \tag{3.4}$$

for the bulk operator $\mathcal{O}_J$ defined in (3.3). The relevant CFT data entering this relation—the bulk one-point function and the bulk—tilt two-point function—were computed in [31, 32]:

$$a_{\mathcal{O}_J} = \frac{\sqrt{\lambda J}}{2^{\frac{J}{2}+1} N} \frac{I_J(\sqrt{\lambda})}{I_1(\sqrt{\lambda})} \,, \tag{3.5}$$

$$b_{\mathcal{O}_J \hat{t}} = \frac{\sqrt{\lambda J^3}}{2^{\frac{J}{2}+1} \pi N} \frac{I_J(\sqrt{\lambda})}{I_1(\sqrt{\lambda})} \,, \tag{3.6}$$

with N the rank of $SU(N)$ gauge group in $\mathcal{N} = 4$ super Yang-Mills. It is straightforward to verify that the relation holds exactly: with the additional factor of J/π, the two coefficients match identically at all orders in the 't Hooft coupling λ.

For later use, we expand $b_{\mathcal{O}_J \hat{t}}$ and $a_{\mathcal{O}_J} C_{\hat{t}}$ in both the weak- and strong-coupling limits to the first few orders:

$$b_{\mathcal{O}_J \hat{t}} = \frac{1}{N} \begin{cases} \frac{2^{-3J/2} J^{3/2}}{\pi \Gamma(J+1)} \lambda^{J/2} - \frac{2^{-3J/2-3} J^{3/2}(J-1)}{\pi (J+1)\Gamma(J+1)} \lambda^{J/2+1} + O(\lambda^{J/2+2}) \,, \\ \frac{J^{3/2}}{\pi 2^{J/2+1}} \sqrt{\lambda} - \frac{J^{3/2}(J^2-1)}{\pi 2^{J/2+2}} + O(\lambda^{-1/2}) \,. \end{cases} \tag{3.7}$$

$$a_{\mathcal{O}_J} C_{\hat{t}} = \frac{1}{N} \begin{cases} \frac{2^{-\frac{3}{2}J-3}\sqrt{J}}{\pi^2 \Gamma(J+1)} \lambda^{J/2+1} + \frac{2^{-\frac{3J}{2}-5}(1-2J)\sqrt{J}}{3\pi^2 (J+1)\Gamma(J+1)} \lambda^{J/2+2} + O(\lambda^{J/2+3}) \,, \\ \frac{2^{-\frac{J}{2}-2}\sqrt{J}}{\pi^2} \lambda - \frac{2^{-\frac{J}{2}-3}\sqrt{J}\left(J^2+2\right)}{\pi^2} \sqrt{\lambda} + O((\lambda^{-1/2})^0) \,. \end{cases} \tag{3.8}$$

3.2 $\int \langle \mathcal{O}\hat{t}\hat{t} \rangle \sim \langle \mathcal{O}\hat{t} \rangle$

In this section, our analysis is restricted to the leading and next-to-leading orders. Apart from the coefficients $a_{\mathcal{O}_J}$ and $b_{\mathcal{O}_J \hat{t}}$ discussed in the previous section, we start by remaining agnostic about any additional CFT data.

Focusing on the case $p = 1$, we may rewrite (2.49) as

$$\begin{aligned} &\langle \mathcal{O}_J(u, x_0)(\hat{u}_1 \cdot t)(\tau_1)(\hat{u}_2 \cdot t)(\tau_2) \rangle \\ &= \frac{1}{N} \frac{(u \cdot \theta)^{J-2}}{|x_{0\perp}|^{\Delta_{\mathcal{O}_J}-2}} \left(\frac{(u \cdot \hat{u}_1)(u \cdot \hat{u}_2)}{|x_{01}|^2 |x_{02}|^2} F_1(\chi) + \frac{(u \cdot \theta)^2 (\hat{u}_1 \cdot \hat{u}_2)}{|x_{0\perp}|^2 |\tau_{12}|^2} F_2(\chi) \right) \,, \end{aligned} \tag{3.9}$$

where the cross-ratio is defined in (2.50) $\chi = \frac{|x_{0\perp}|^2 |\tau_{12}|^2}{|x_{01}|^2 |x_{02}|^2}$. Note that here we consider the full correlator rather than the connected part. Their relation is given by $F_{1c} = F_1$ and $F_{2c} = F_2 - a_{\mathcal{O}_J} C_{\hat{t}}$, as discussed in Section 2.2.2.

In particular, the pinching and splitting limits introduced in [32] provide non-perturbative constraints on these functions. The pinching limit corresponds to taking $\tau_1 \to \tau_2$, where the defect insertions approach each other and the correlator reduces to a lower-point function along the Wilson loop. A different limit comes from considering a large separation between the bulk operator and the defect operators. In the correlator expressed in R-symmetry channels, this corresponds to the limit $x_\perp \to \infty$. In this case, the bulk–defect–defect correlator factorizes into a product of a two-point defect correlator and a one-point function of the bulk operator. Applying these limits leads to the exact relations

$$F_1(0) = b_{\mathcal{O}_J\hat{T}}\lambda_{\hat{t}\hat{t}\hat{T}}, \qquad F_2(0) = a_{\mathcal{O}_J}C_{\hat{t}}\,. \tag{3.10}$$

In $\mathcal{N}=4$ super Yang-Mills, the 1/2 BPS Wilson line is known to possess a topological sector [31, 33–36], consisting of correlators of 1/2 BPS operators that become effectively topological when the spacetime and R-symmetry polarizations are aligned. In this limit, the correlators lose their dependence on continuous spacetime separations and depend only on discrete data such as operator ordering or internal symmetry structure. In our setup, we focus on the bulk–tilt–tilt correlator and assume that when the polarization vectors u and $\hat{u}_{1,2}$ are chosen as appropriate functions of the cross ratio χ, the correlator reduces to its topological form. We parametrize it as

$$\mathbb{F}_J = \kappa_1(\chi)F_1(\chi) + \kappa_2(\chi)F_2(\chi)\,, \tag{3.11}$$

where $\kappa_1(\chi)$ and $\kappa_2(\chi)$ are, a priori, unknown functions of χ. This structure reflects the general decomposition of the correlator into R-symmetry channels. However, for the specific definitions of F_1 and F_2 given in (3.9), both functions are expected to be constants, as argued in [32]. To fix the overall normalization, we set $\kappa_1 = 1$.

This topological sector admits an equivalent interpretation in terms of the OPE of defect operators. In this description, the exchanged operators are 1/2-BPS and hence protected from quantum corrections. The correlator can then be expressed as

$$\mathbb{F}_J = \sigma_{1J}b_{\mathcal{O}_J\hat{T}}\lambda_{\hat{t}\hat{t}\hat{T}} + \sigma_{2J}a_{\mathcal{O}_J}C_{\hat{t}}\,, \tag{3.12}$$

where σ_{1J} and σ_{2J} are numerical coefficients that may depend on the bulk operator dimension J. Combining this representation with the general form above, one can write

$$F_1(\chi) = \sigma_{1J}b_{\mathcal{O}_J\hat{T}}\lambda_{\hat{t}\hat{t}\hat{T}} + \sigma_{2J}a_{\mathcal{O}_J}C_{\hat{t}} - \kappa_2F_2(\chi)\,. \tag{3.13}$$

3.2.1 Weak coupling

We now focus on the weak-coupling regime. At leading order in $\lambda^{J/2}$, the function F_2 does not contribute. At next-to-leading order, it receives contributions only from disconnected Feynman diagrams, as illustrated in Figure 1. The left diagram represents planar disconnected contributions that survive in the large-N limit, while the right diagram corresponds to non-planar contributions that vanish in this limit. Although these diagrams are disconnected, F_2 is not strictly constant: the integration over the positions of the Wilson line insertions must be treated carefully, as the integration region and ordering of the propagators can introduce a nontrivial dependence on the cross ratio χ.

Figure 1: Disconnected Feynman diagrams contributing to the $\langle \mathcal{O}_J tt \rangle$ correlators at weak coupling. The ellipsis indicates the $J-3$ propagators connecting the bulk operator to the Wilson line that have been omitted for clarity.

Following [32], the planar contribution from the left diagram evaluates to[3]

$$F_2(\chi) = \lambda^{J/2+1} c_J \sum_{\pm} \left(\pi \pm 2 \arctan \sqrt{\frac{1-\chi}{\chi}} \right)^J + O(\lambda^{J/2+2}) \,, \tag{3.14}$$

where c_J is a constant that can be fixed using the pinching and splitting limits (3.10). Inserting the data from (3.8), we obtain

$$c_J = \frac{2^{-\frac{5}{2}J-3}\sqrt{J}}{\pi^{J+2}\Gamma(J+1)} \,. \tag{3.15}$$

However, we emphasize that the disconnected diagrams are not equivalent to the disconnected correlators. As discussed in (2.54) and in Appendix B, the disconnected correlators are defined by subtracting the factorized contribution from the full correlator. In the present case, this corresponds to removing the product of the bulk one-point function and the tilt two-point function. Explicitly, this yields (2.55). Plugging it to the identity (2.58), it becomes

$$\begin{aligned} &2\frac{b_{\mathcal{O}_J\hat{\phi}}\lambda_{\hat{t}\hat{t}\hat{\phi}}}{\gamma_{\hat{\phi}}} + \int_0^{1/2} d\chi \left(\frac{F_{2c}(\chi)}{\chi^{3/2}(1-\chi)^{1/2}} - \frac{b_{\mathcal{O}_J\hat{\phi}}\lambda_{tt\hat{\phi}}}{\chi^{1-\gamma_{\hat{\phi}}/2}(1-\chi)^{1+\gamma_{\hat{\phi}}/2}} \right) \\ &+ \int_{1/2}^1 d\chi \frac{F_{2c}(\chi)}{\chi^{3/2}(1-\chi)^{1/2}} = -\frac{2^{-3J/2}J^{5/2}}{\pi\Gamma(J+1)}\lambda^{J/2} + \frac{2^{-3J/2-3}J^{5/2}(J-1)}{\pi(J+1)\Gamma(J+1)}\lambda^{J/2+1} \,. \end{aligned} \tag{3.16}$$

Note again that the $\frac{1}{\chi^{1-\gamma_{\hat{\phi}}/2}(1-\chi)^{1+\gamma_{\hat{\phi}}/2}}$ term must be expanded like (2.59). Here, the left-hand side receives contributions from both the contact term and the integrated part of the disconnected correlator. However, at leading order in $\lambda^{J/2}$, the integral is subleading, and the only contribution comes from the contact term. Using

$$\lambda_{\hat{t}\hat{t}\hat{\phi}} = \frac{\lambda^{3/2}}{16\sqrt{2}\pi^3} - \frac{(18+\pi^2)\,\lambda^{5/2}}{768\sqrt{2}\pi^5}, \qquad \gamma_{\hat{\phi}} = \frac{\lambda}{4\pi^2} - \frac{\lambda^2}{16\pi^4} + \frac{128 - \frac{56\pi^4}{45}}{4096\pi^6}\lambda^3 + O(\lambda^4) \,, \tag{3.17}$$

[3]As discussed later in this work, several results in [32] appear to be in tension with those implied by the integral identities. Here, however, we choose to start with this particular result.

we find

$$b_{\mathcal{O}_J\hat{\phi}} = -\frac{2^{-3(J-1)/2}J^{5/2}}{\Gamma(J+1)}\lambda^{(J-1)/2} + O(\lambda^{(J+1)/2})\,. \tag{3.18}$$

There is an alternative way to compute $b_{\mathcal{O}_J\phi}$ at this order. From the analysis following (2.57), we have

$$(b_{\mathcal{O}_J\hat{\phi}}\lambda_{\hat{t}\hat{t}\hat{\phi}})|_{\lambda^{J/2+1}} = \lim_{\tau\to 0}\frac{1}{\tau}F_{2c}\left(\frac{\tau^2}{1+\tau^2}\right)\bigg|_{\lambda^{J/2+1}} = -\frac{2^{-3(J-1)/2}J^{3/2}}{\Gamma(J+1)}\,. \tag{3.19}$$

However, this result differs by a factor of $1/J$ from the expected value (3.18), indicating a mismatch. This discrepancy suggests that the expression F_2 obtained in (3.14) is problematic.

From the Feynman diagram analysis shown in Figure 2, we see that, in contrast to F_2, the F_1 channel contributes already at order $\lambda^{J/2}$ as a constant. This is consistent with (3.13), since F_2 vanishes at leading order. In particular, this fixes that $\sigma_{1J} = 1$. Using the pinching and splitting limits (3.10), we can fix this constant, giving

$$F_1(\chi) = \lambda^{J/2}(b_{\mathcal{O}_J\hat{T}}\lambda_{\hat{t}\hat{t}\hat{T}})|_{\lambda^{J/2}} + O(\lambda^{J/2+1})\,. \tag{3.20}$$

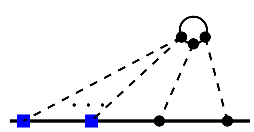


Figure 2: Feynman diagram contributing at leading order in the F_1 channel.

From (2.63) for $p=1$, we have

$$\pi\left(\lambda_{\hat{t}\hat{t}\hat{T}}b_{\mathcal{O}_J\hat{T}}\right)|_{\text{leading}} = J(J-1)b_{\mathcal{O}_Jt}|_{\text{leading}}\,. \tag{3.21}$$

which leads to the explicit expression

$$b_{\mathcal{O}_J\hat{T}}\lambda_{\hat{t}\hat{t}\hat{T}} = \frac{2^{-3J/2}J^{3/2}}{\pi^2\Gamma(J-1)}\lambda^{J/2} + O(\lambda^{J/2+1})\,. \tag{3.22}$$

At the next-to-leading order, the Feynman diagrams contributing to the F_1 channel become significantly more involved. However, we can bypass the explicit diagrammatic computation by making use of (3.13), which provides a shortcut to determine F_1 at this order without evaluating the full set of diagrams. Specifically, Substituting (3.13) into (2.52) for $p=1$, we obtain

$$\pi(b_{\mathcal{O}_J\hat{T}}\lambda_{\hat{t}\hat{t}\hat{T}} + \sigma_{2J}a_{\mathcal{O}_J}C_{\hat{t}}) - \frac{\kappa_2}{N}\int_0^1\frac{d\chi\,F_2(\chi)}{(\chi(1-\chi))^{1/2}} = J(J-1)b_{\mathcal{O}_Jt}\,. \tag{3.23}$$

so at order $\lambda^{J/2+1}$, it allows us to get a relation

$$\kappa_2 = \frac{1}{2}\left((J+1)\sigma_{2J} + J^2(1-J)^2\right) . \tag{3.24}$$

However, we find that the explicit results of [32] do not satisfy this relation. This mismatch may stem from subtleties in the definition of the topological limit, the normalization adopted in their perturbative computation, or issues associated with the planar limit. The discrepancy already appears at leading order. For instance, compared to (3.22), their result reads

$$(b_{\mathcal{O}_J\hat{T}}\lambda_{\hat{t}\hat{t}\hat{T}})' = \frac{2^{-3J/2}J^{1/2}}{\pi^2\Gamma(J-1)}\lambda^{J/2} + O(\lambda^{J/2+1}) . \tag{3.25}$$

which differs by an additional factor of $1/J$.

3.2.2 Strong coupling

At strong coupling, there is, as of now, relatively little that can be said in general terms. The leading and next-to-leading correlators were computed in [32]. However, if we assume that $F_1(\chi)$ remains constant at both leading and next-to-leading orders as claimed in that paper, then using (2.52) we obtain

$$F_1(\chi) = \frac{(J-1)J^{5/2}}{\pi^2 2^{J/2+1}}\sqrt{\lambda} - \frac{J^{5/2}(J-1)^2(J+1)}{\pi^2 2^{J/2+2}} + O(\lambda^{-1/2}) . \tag{3.26}$$

For $J=2$, this simplifies to

$$F_1(\chi) = \frac{\sqrt{2}}{\pi^2}\sqrt{\lambda} - \frac{3}{\sqrt{2}\pi^2} + O(\lambda^{-1/2}) . \tag{3.27}$$

Comparing this with the results of [32]—where we have accounted for the canonical normalization of the tilt operators, as well as the additional factor of -2 arising from our different definition of F_2[4]—we find

$$F_1 = -\frac{3}{\sqrt{2}\pi^2} + O(\lambda^{-1/2}) , \tag{3.28}$$

which agrees with our expression at next-to-leading order but differs at leading order.

Moreover, at both leading and next-to-leading orders, $F_2(\chi)$ cannot be constant, contrary to the claim in [32]. If $F_2(\chi)$ were constant, it would violate (2.53), since in this regime there are no contact terms analogous to those appearing at weak coupling, and the integral relation (2.53) should therefore hold exactly without any subtleties or modifications. Furthermore, this observation implies that $F_1(\chi)$ cannot remain constant either, as the two channels are linked through the topological sector, see (3.11). Taken together, these results indicate that,

[4]We thank Daniele Artico and Julien Barrat for updating F_1 and F_2 in [32] at next-to-leading order.

at both weak and strong coupling, the bulk-defect-defect correlators calculated in [32] are inconsistent with the integral identities. We anticipate that our identities will provide a robust framework for checking and revising existing correlators, leaving a complete reassessment for future work.

4 Example: Magnetic lines in $O(N)$

The bulk $O(N)$ model in Euclidean $d = 4 - \varepsilon$ dimensions consists of N real scalar fields ϕ_a, $a = 1, \dots, N$, governed by the action

$$S_{O(N)} = \int d^d x \left(\frac{1}{2} \partial_\mu \phi_a \partial^\mu \phi_a + \frac{1}{4!} \lambda_0 \left(\phi_a \phi_a \right)^2 \right) . \tag{4.1}$$

The magnetic line, taken to lie along $x^\mu(\tau) = (\tau, \mathbf{0})$, is given by deforming the bulk action,

$$S_{O(N)} \to S_{O(N)} + h_0 \int_{-\infty}^{\infty} d\tau \phi_N(\tau) , \tag{4.2}$$

where we use the $O(N)$ symmetry to take ϕ_N as the direction which couples to the line. This breaks the bulk $O(N)$ symmetry to $O(N-1)$. When working perturbatively in ε, this theory exhibits an RG flow to an infrared fixed point which, to linear order, is characterized by

$$\frac{\lambda_*}{(4\pi)^2} = \frac{3\varepsilon}{N+8} + O(\varepsilon^2) , \quad h_*^2 = N + 8 + \frac{4N^2 + 45N + 170}{2(N+8)} \varepsilon + O(\varepsilon^2) . \tag{4.3}$$

A consequence of the $O(N) \to O(N-1)$ symmetry breaking is the modification of the conservation equations for the broken $O(N)$ currents $J^\mu_{\imath N}$, where $\imath = 1, \dots, N-1$. These equations now read as

$$\partial_\mu J^\mu_{\imath N}(\tau, x_\perp) = \delta(x_\perp) \hat{t}_\imath(\tau) . \tag{4.4}$$

The defect operator $\hat{t}$ is the tilt operator and has protected scaling dimension $\Delta_{\hat{t}} = 1$. Moreover, the tilt operator inherits a canonical normalization such that

$$\langle \hat{t}_\imath(\tau) \hat{t}_\jmath(0) \rangle = \delta_{\imath\jmath} \frac{\mathcal{N}_{\hat{t}}^2}{\tau^2} , \quad \mathcal{N}_{\hat{t}}^2 = \frac{h_*^2}{4\pi^2} \left(1 - \frac{\varepsilon}{2} + O(\varepsilon^2) \right) . \tag{4.5}$$

In the following sections we demonstrate a variety of integral identities each with a single bulk operator insertion, a tilt operator inserted on the defect, and either zero, one or two other defect operators. We primarily consider bulk operators with dimensions close to one and two and defect operators with dimensions close to one. All operators are unit-normalized with the exception of the tilt operator.

4.1 $\int \langle \mathcal{O}\hat{t}\rangle \sim \langle \mathcal{O}\rangle$

The first concrete example that we consider is the relation between bulk one-point functions and bulk-defect two-point functions. The identity that we find corresponds to (2.38) and reads

$$\int d\tau_1 \langle \mathcal{O}_{a_1\ldots a_n}(x_0)\hat{t}_\imath(\tau_1)\rangle = \langle (Q_\imath \mathcal{O}_{a_1\ldots a_n})(x_0)\rangle \,, \tag{4.6}$$

where $\mathcal{O}_{a_1\ldots a_n}$ is a bulk operator that transforms in some irreducible representation of $O(N)$ and $Q_\imath \equiv Q_{\imath N}$ is the broken Noether charge associated to the broken current $J^\mu_{\imath N}$. The action of $Q_\imath$ on a fundamental field ϕ_a is given by

$$Q_\imath \phi_a = \delta_{\imath a}\phi_N - \delta_{Na}\phi_\imath \,. \tag{4.7}$$

It turns out to be straightforward to derive the consequences of this identity in general, largely thanks to the fact that there are no cross-ratios in this configuration, and the correlation functions are completely fixed by conformal symmetry. Indeed, up to tensor structures, the forms of the bulk one-point function and the bulk-defect two point function are

$$\langle \mathcal{O}_{a_1\ldots a_n}(x_0)\rangle = \mathcal{T}_{a_1\ldots a_n}\frac{a_{\mathcal{O}}}{|x_{0\perp}|^{\Delta_{\mathcal{O}}}} \,, \tag{4.8}$$

$$\langle \mathcal{O}(x_0)\hat{t}_\imath(\tau_1)\rangle = \mathcal{T}_{a_1\ldots a_n;\imath}\frac{b_{\mathcal{O}\hat{\mathcal{O}}'}}{(|x_{0\perp}|^2+\tau_{01}^2)^{\Delta_{\hat{\mathcal{O}}'}}|x_{0\perp}|^{\Delta_{\mathcal{O}}-\Delta_{\hat{\mathcal{O}}'}}} . \tag{4.9}$$

The right hand side of (4.6) can be written as

$$\langle Q_\imath \mathcal{O}_{a_1\ldots a_n}(x_0)\rangle = \sum_{k=1}^{n}[Q_\imath]_{a_k c}\langle \mathcal{O}_{a_1\ldots a_{k-1}c a_{k+1}\ldots a_n}(x_0)\rangle, \tag{4.10}$$

where $[Q_\imath]_{a_k c} = \delta_{\imath a_k}\delta_{Nc} - \delta_{Na_k}\delta_{\imath c}$. Since $\mathcal{O}$ transforms in an *irreducible* representation of $O(N)$, this is just a sum of one-point functions of $\mathcal{O}$ and the overall tensorial structure must match that of the left-hand side of (4.6). This means that upon plugging (4.8) and (4.9) into (4.6), we have

$$\sum_{k=1}^{n}[Q_\imath]_{a_k c}\mathcal{T}_{a_1\ldots c\ldots a_n}\frac{a_{\mathcal{O}}}{|x_{0\perp}|^{\Delta_{\mathcal{O}}}} = \int_{-\infty}^{\infty} d\tau_1 \frac{\mathcal{T}_{a_1\ldots a_n;\imath}b_{\mathcal{O}\hat{t}}}{(|x_{0\perp}|^2+(\tau_0-\tau_1)^2)|x_{0\perp}|^{\Delta_{\mathcal{O}}-1}} \tag{4.11}$$

$$= \mathcal{T}_{a_1\ldots a_n;\imath}\frac{\pi b_{\mathcal{O}\hat{t}}}{|x_{0\perp}|^{\Delta_{\mathcal{O}}}} \,. \tag{4.12}$$

From this, we find the relation

$$\sum_{k=1}^{n}[Q_\imath]_{a_k c}\mathcal{T}_{a_1\ldots c\ldots a_n}a_{\mathcal{O}} = \mathcal{T}_{a_1\ldots a_n;\imath}\pi b_{\mathcal{O}\hat{t}} \,. \tag{4.13}$$

When the bulk operator transforms in a traceless symmetric representation, then the above relation can be contracted with null polarization vectors. Upon doing so, we recover (2.45).

We can take $\mathcal{O} = \phi_a$. Then $b_{\phi\hat{t}}$ has been calculated in [37] to be

$$b_{\phi\hat{t}} = \mathcal{N}_{\hat{t}} \left(1 - \frac{1 - \log 2}{2}\varepsilon + O(\varepsilon^2)\right) . \tag{4.14}$$

Using this in (4.13), we find

$$a_\phi = \frac{\sqrt{N+8}}{2} \left(1 + \left(\log\sqrt{2} + \frac{N^2 - 3N - 22}{4(N+8)^2}\right)\varepsilon\right) , \tag{4.15}$$

which agrees with what is calculated in [37] via Feynman diagrams.

If we instead take $\mathcal{O} = S \equiv Z_S^{-1}\phi_a\phi_a$, then $\langle S(x_1)\hat{t}_\imath(\tau_2)\rangle = 0$ and $Q_\imath S = 0$, which means that (4.6) is trivial and we do not find any constraint.

Finally, we take $\mathcal{O} = T_{ab} \equiv Z_T^{-1}(\phi_a\phi_b - \mathrm{tr})$. From a Feynman diagram calculation, we find

$$b_{T\hat{t}} = -\frac{N+8}{4\sqrt{2}\pi} \left(1 + \left(\frac{N+6}{N+8}\log 2 + \frac{N^2 - 5N - 38}{2(N+8)^2}\right)\varepsilon\right) . \tag{4.16}$$

In this case, (4.13) gives

$$a_T = -\frac{N+8}{4\sqrt{2}} \left(1 + \left(\frac{N+6}{N+8}\log 2 + \frac{N^2 - 5N - 38}{2(N+8)^2}\right)\varepsilon\right) , \tag{4.17}$$

which also agrees with results in [37].

4.2 $\int\langle\mathcal{O}\hat{\mathcal{O}}\hat{t}\rangle \sim \langle\mathcal{O}\hat{\mathcal{O}}\rangle$

A general bulk-defect-defect correlator in this theory has the form

$$\langle\mathcal{O}_1(x_1)\hat{\mathcal{O}}_2(\tau_2)\hat{\mathcal{O}}_3(\tau_3)\rangle = \frac{|x_{1\perp}|^{\hat{\Delta}_2+\hat{\Delta}_3-\Delta_1}}{|x_{12}|^{2\hat{\Delta}_2}|x_{13}|^{2\hat{\Delta}_3}} F_{\mathcal{O}_1\hat{\mathcal{O}}_2\hat{\mathcal{O}}_3}(\chi), \tag{4.18}$$

where $F_{\mathcal{O}_1\hat{\mathcal{O}}_2\hat{\mathcal{O}}_3}$ is some function of the cross-ratio,

$$\chi = \frac{x_{1\perp}^2\tau_{23}^2}{x_{12}^2x_{13}^2} . \tag{4.19}$$

4.2.1 Bulk Operator Dimension ~ 1

Let us now consider the integral identity that involves integrating $\langle\phi_a\hat{t}_\imath\hat{t}_\jmath\rangle_c$. This corresponds to calculations in section 2.2.2. We use conformal symmetry to place the operators at $x_1 = (0,1)$, $\tau_2 = 0$ and $\tau_3 \equiv \tau$ is to be integrated along the line. The cross-ratio for this configuration is $\chi = \frac{\tau^2}{1+\tau^2}$ and

$$\langle\phi_a(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = \delta_{aN}\delta_{\imath\jmath}\varepsilon\frac{\sqrt{N+8}}{16\pi^2}\frac{\pi^2 - 4\arccos^2\sqrt{\frac{\tau^2}{1+\tau^2}}}{\tau^2} . \tag{4.20}$$

Since this function is even in τ, we can replace $\int_{-\infty}^{\infty} d\tau = 2\int_0^{\infty} d\tau$. Moreover, we have the following behaviours,

$$\langle \phi_a(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = \delta_{aN}\delta_{\imath\jmath}\varepsilon\frac{\sqrt{N+8}}{4\pi\tau} + O(\tau^0)\,, \quad \tau\to 0\,, \tag{4.21}$$

$$\langle \phi_a(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = O\left(\frac{1}{\tau^2}\right)\,, \quad \tau\to\infty\,. \tag{4.22}$$

We therefore isolate the divergent part of the integral by splitting the integral into two parts, one over $\tau\in(0,1)$ and the other over $\tau\in(1,\infty)$, as is done in (2.57). The first part requires careful handling of the divergence. Subtracting the pole in (4.21) from the connected three-point function leaves an expression which is integrable over $\tau\in(0,1)$. The expression in (4.20) is integrable over $\tau\in(1,\infty)$, which means that we have

$$\int_0^1 d\tau\left(\langle \phi_a(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c - \delta_{aN}\delta_{\imath\jmath}\frac{\varepsilon\sqrt{N+8}}{4\pi\tau}\right) + \int_1^\infty d\tau\langle \phi_a(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = \varepsilon\sqrt{N+8}\frac{1-\log 2}{4\pi}\,. \tag{4.23}$$

The contribution from the pole needs to be restored, and we do that by analyzing the $\hat{t}\times\hat{t}$ OPE which to leading order takes the form

$$\hat{t}_i(0)\times\hat{t}_\jmath(\tau) = \delta_{\imath\jmath}\frac{\mathcal{N}_{\hat{t}}^2\lambda_{\hat{t}\hat{t}\hat{\phi}}}{|\tau|^{2-\Delta_{\hat{\phi}}}}\hat{\phi}(0)+\dots\,, \tag{4.24}$$

where

$$\Delta_{\hat{\phi}} = 1+\varepsilon-\frac{3N^2+49N+194}{2(N+8)^2}\varepsilon^2+O(\varepsilon^3)\,, \tag{4.25}$$

$$\lambda_{\hat{t}\hat{t}\hat{\phi}} = \frac{\pi}{\sqrt{N+8}}\varepsilon-\frac{(9N^2+127N+494)\pi}{4(N+8)^{5/2}}\varepsilon^2+O(\varepsilon^3)\,. \tag{4.26}$$

The contribution from the identity operator to the OPE has been ignored since it is disconnected. This can be substituted into the three-point function and it can then be verified that

$$\frac{\mathcal{N}_{\hat{t}}^2\lambda_{\hat{t}\hat{t}\hat{\phi}}b_{\phi\hat{\phi}}}{\tau^{2-\Delta_{\hat{\phi}}}} = \varepsilon\frac{\sqrt{N+8}}{4\pi\tau}+O(\varepsilon^2)\,, \tag{4.27}$$

which matches the pole in (4.21), showing that this singularity comes from the $\hat{\phi}$ contribution to the $t\times t$ OPE. The expression on the left-hand side of (4.27) can be integrated over $\tau\in(0,1)$ and gives

$$\int_0^1 d\tau\frac{\mathcal{N}_{\hat{t}}^2\lambda_{\hat{t}\hat{t}\hat{\phi}}b_{\phi\hat{\phi}}}{\tau^{2-\Delta_{\hat{\phi}}}} = \frac{\sqrt{N+8}}{4\pi}\left(1+\varepsilon\left(\frac{3}{2}\log 2-\frac{3N^2+67N+278}{4(N+8)^2}\right)\right)\,. \tag{4.28}$$

Combining (4.23) and (4.28), and multiplying by a factor of 2 to account for integrating over the whole real line, we find

$$\int_{-\infty}^{\infty} \langle \phi_a(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = \delta_{aN}\delta_{\imath\jmath}\frac{\sqrt{N+8}}{2\pi}\left(1+\varepsilon\left(\frac{N^2-3N-22}{4(N+8)^2}+\frac{1}{2}\log 2\right)\right) , \tag{4.29}$$

which is precisely $\delta_{aN}\delta_{\imath\jmath}b_{\phi\hat{t}}$ to $O(\varepsilon)$.

We can also integrate $\langle \phi_a\hat{\phi}\hat{t}_\imath\rangle_c$ in a similar way. The correlation function and pole are

$$\langle \phi_a(0,1)\hat{\phi}(0)\hat{t}_\imath(\tau)\rangle_c = \delta_{a\imath}\frac{\varepsilon}{8\pi}\frac{\pi^2-4\arccos^2\sqrt{\frac{\tau^2}{1+\tau^2}}}{\tau^2} , \tag{4.30}$$

$$= \delta_{a\imath}\frac{\varepsilon}{2\tau}+O(\tau^0) , \quad \tau\to 0 . \tag{4.31}$$

As before, we exploit the fact that the correlation function is even in τ, so we integrate over the positive real line and then multiply by a factor of 2 in order to get the final result. Proceeding as we did in the previous example, we find

$$\int_0^1 d\tau\left(\langle \phi_a(0,1)\hat{\phi}(0)\hat{t}_i(\tau)\rangle_c-\delta_{a\imath}\frac{\varepsilon}{2\tau}\right)+\int_1^{\infty} d\tau\langle \phi_a(0,1)\hat{\phi}(0)\hat{t}_i(\tau)\rangle_c = \delta_{a\imath}\varepsilon\frac{1-\log 2}{2} . \tag{4.32}$$

The relevant OPE for understanding the pole is $\hat{\phi}\times\hat{t}$, which to leading order is

$$\hat{\phi}(0)\times\hat{t}_\imath(\tau) = \frac{\lambda_{\hat{t}\hat{t}\hat{\phi}}}{|\tau|^{\Delta_{\hat{\phi}}}}\hat{t}_\imath(0)+\dots . \tag{4.33}$$

This is the term that produces the pole, since

$$\frac{\lambda_{\hat{t}\hat{t}\hat{\phi}}b_{\phi\hat{t}}}{\tau^{\Delta_{\hat{\phi}}}} = \frac{\varepsilon}{2\tau}+O(\varepsilon^2) . \tag{4.34}$$

In order to integrate the left-hand side of (4.34), we use the formula, valid in the domain $\tau\in(0,1)$,

$$\frac{\varepsilon}{\tau^{1+\varepsilon\gamma}} = -\frac{1}{\gamma}\delta(\tau)+\frac{\varepsilon}{\tau}+\dots . \tag{4.35}$$

When integrating over $\tau\in(0,1)$, the finite contribution comes from the term involving the delta function. Combining this with (4.32), we find

$$\int_{-\infty}^{\infty} d\tau\langle \phi_a(0,1)\hat{\phi}(0)\hat{t}_\imath(\tau)\rangle_c = -\delta_{a\imath}\left(1-\frac{3-3\log 2}{2}\varepsilon\right) = -\delta_{a\imath}b_{\phi\hat{\phi}} , \tag{4.36}$$

as expected.

4.2.2 Bulk Operator Dimension ~ 2

Of the two bulk operators at this dimension, we first consider the singlet S. There is only one non-zero three point function involving S and defect operators with dimensions close to one that can be integrated in our framework, $\langle S\hat{t}_\imath\hat{t}_\jmath\rangle_c$. The correlation function and pole are

$$\begin{aligned}\langle S(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c =&\delta_{\imath\jmath}\frac{N+8}{2\pi^2\sqrt{2N}}\Bigg(\frac{1}{1+\tau^2}+\varepsilon\Bigg(-\frac{\pi^2-\arccos^2\sqrt{\frac{\tau^2}{1+\tau^2}}}{8\tau^2} \qquad (4.37)\\ &+\frac{1}{1+\tau^2}\left(-\frac{13N+38}{2(N+8)^2}+\log 2+\frac{N+2}{2(N+8)}\log\frac{\tau^2}{1+\tau^2}\right)\Bigg)\Bigg)\,,\\ =&-\delta_{\imath\jmath}\varepsilon\frac{N+8}{4\pi\sqrt{2N}\tau}+O(\tau^0)\,. \qquad (4.38)\end{aligned}$$

We proceed in the familiar way by splitting up the integration region and removing the pole in the appropriate term,

$$\begin{aligned}\int_0^1 d\tau\left(\langle S(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c+\delta_{\imath\jmath}\varepsilon\frac{N+8}{4\pi\sqrt{2N}\tau}\right)+\int_1^\infty d\tau\langle S(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c&\\ =\delta_{\imath\jmath}\frac{N+8}{4\pi\sqrt{2N}}\left(1+\varepsilon\left(\frac{N+14}{N+8}\log 2-\frac{2N^2+45N+166}{2(N+8)^2}\right)\right)\,.&\qquad (4.39)\end{aligned}$$

The pole comes from the $\hat{\phi}$ contribution to the $\hat{t}\times\hat{t}$ OPE. Indeed, after substituting this contribution into the three-point function, we find

$$\frac{\mathcal{N}_{\hat{t}}^2\lambda_{\hat{t}\hat{t}\hat{\phi}}b_{S\hat{\phi}}}{\tau^{2-\Delta_{\hat{\phi}}}}=-\varepsilon\frac{N+8}{4\pi\sqrt{2N}\tau}+O(\varepsilon^2)\,. \qquad (4.40)$$

To determine the contribution from the pole to the integral, we integrate the left-hand side of (4.40) finding precisely the expression in (4.39) with an extra overall minus sign. Combining these two expressions gives us the expected result

$$\int_{-\infty}^{\infty}d\tau\langle S(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c=0\,. \qquad (4.41)$$

As for the traceless-symmetric operator T_{ab}, we first consider $\langle T_{ab}\hat{\phi}\hat{t}_\imath\rangle_c$. The correlation function is

$$\begin{aligned}\langle T_{ab}(0,1)\hat{\phi}(0)\hat{t}_\imath(\tau)\rangle_c=\mathcal{T}^{(1)}_{ab;\imath}\frac{\sqrt{N+8}}{2\sqrt{2}\pi}\Bigg(\frac{1}{1+\tau^2}+\varepsilon\Bigg(\frac{2}{1+\tau^2}\log 2&\\ -\frac{5N^2+103N+438}{4(N+8)^2(1+\tau^2)}+\frac{1}{(N+8)(1+\tau^2)}\log\frac{\tau^2}{1+\tau^2}-\frac{\pi^2-4\arccos^2\sqrt{\frac{\tau^2}{1+\tau^2}}}{8\tau^2}\Bigg)\Bigg)\,,&\qquad (4.42)\end{aligned}$$

where

$$\mathcal{T}^{(1)}_{ab;\imath}=\delta_{a\imath}\delta_{bN}+\delta_{b\imath}\delta_{aN}\,, \qquad (4.43)$$

and the behaviour for small τ is

$$\langle T_{ab}(0,1)\hat{\phi}(0)\hat{t}_\imath(\tau)\rangle_c = -\mathcal{T}^{(1)}_{ab;\imath}\frac{\varepsilon\sqrt{N+8}}{4\sqrt{2}\tau} + O(\tau^0)\,. \tag{4.44}$$

Integrating in the usual way, we find

$$\begin{aligned}\int_0^1 d\tau \left(\langle T_{ab}(0,1)\hat{\phi}(0)\hat{t}_\imath(\tau)\rangle_c + \mathcal{T}^{(1)}_{ab;\imath}\frac{\varepsilon\sqrt{N+8}}{4\sqrt{2}\tau}\right) + \int_1^\infty d\tau \langle T_{ab}(0,1)\hat{\phi}(0)\hat{t}_\imath(\tau)\rangle_c \\ = \mathcal{T}^{(1)}_{ab;\imath}\frac{\sqrt{N+8}}{4\sqrt{2}}\left(1+\varepsilon\left(\frac{3N+22}{N+8}\log 2 - \frac{9N^2+167N+694}{4(N+8)^2}\right)\right)\,.\end{aligned} \tag{4.45}$$

The pole can be seen to come from the $\hat{t}_\imath$ contribution to the $\hat{\phi}\times\hat{t}_\imath$ OPE, since we have

$$\frac{\lambda_{\hat{t}\hat{t}\hat{\phi}}b_{T\hat{t}}\mathcal{T}^{(1)}_{ab;\imath}}{\tau^{\Delta_{\hat{\phi}}}} = -\mathcal{T}^{(1)}_{ab;\imath}\frac{\varepsilon\sqrt{N+8}}{4\sqrt{2}\tau} + O(\varepsilon^2)\,, \tag{4.46}$$

where $b_{T\hat{t}}$ is given in (4.16) and we take the tensor structure appearing in the $\langle T_{ab}\hat{t}_\imath\rangle$ two-point function to be $\mathcal{T}^{(1)}_{ab;\imath}$. As before, we use (4.35) in order to integrate the left-hand side of (4.46). The finite contribution comes from the delta function term. Including this with (4.45) gives

$$\begin{aligned}\int_{-\infty}^\infty d\tau \langle T_{ab}(0,1)\hat{\phi}(0)\hat{t}_\imath(\tau)\rangle_c = \mathcal{T}^{(1)}_{ab;\imath}\sqrt{\frac{N+8}{2}}\Bigg(1 \\ +\varepsilon\left(\frac{2(N+7)}{N+8}\log 2 - \frac{5N^2+103N+438}{4(N+8)^2}\right)\Bigg)\,.\end{aligned} \tag{4.47}$$

This integral identity also involves terms from the action of the broken symmetry generator acting on the bulk operator T_{ab}. These terms are

$$\delta_{a\imath}\langle T_{Nb}(0,1)\hat{\phi}(0)\rangle_c - \delta_{aN}\langle T_{\imath b}(0,1)\hat{\phi}(0)\rangle_c + (a\leftrightarrow b)\,. \tag{4.48}$$

The two-point function coefficient appearing in these terms is calculated to be

$$b_{T\hat{\phi}} = -\sqrt{\frac{N+8}{2}}\left(1+\varepsilon\left(\frac{2(N+7)}{N+8}\log 2 - \frac{5N^2+103N+438}{4(N+8)^2}\right)\right)\,, \tag{4.49}$$

where we have taken the tensor structure appearing in the $\langle T_{ab}\hat{\phi}\rangle$ two-point function to be

$$\mathcal{T}^{(2)}_{ab} = \delta_{aN}\delta_{bN} - \frac{1}{N}\delta_{ab}\,. \tag{4.50}$$

The overall tensor structure appearing in (4.48) is precisely $\mathcal{T}^{(1)}_{ab;\imath}$, which means that the sum of these terms together with (4.47) vanishes, as required.

The final three-point function example is $\langle T_{ab}\hat{t}_\imath\hat{t}_\jmath\rangle_c$,

$$\langle T_{ab}(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = -\mathcal{T}^{(2)}_{ab}\delta_{\imath\jmath}\frac{\varepsilon(N+8)}{16\sqrt{2}\pi^2}\frac{\pi^2-4\arccos^2\sqrt{\frac{\tau^2}{1+\tau^2}}}{\tau^2}$$
$$+\mathcal{T}^{(3)}_{ab;\imath\jmath}\frac{N+8}{2\sqrt{2}\pi^2(1+\tau^2)}\left(1+\varepsilon\left(\log 2+\frac{N^2-5N-38}{2(N+8)^2}+\frac{1}{N+8}\log\frac{\tau^2}{1+\tau^2}\right)\right)\,, \quad (4.51)$$

where

$$\mathcal{T}^{(3)}_{ab;\imath\jmath} = \frac{1}{2}\delta_{a\imath}\delta_{b\jmath}+\frac{1}{2}\delta_{a\jmath}\delta_{b\imath}-\frac{1}{N}\delta_{ab}\delta_{\imath\jmath}\,. \quad (4.52)$$

The pole in τ is

$$\langle T_{ab}(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = -\mathcal{T}^{(2)}_{ab}\delta_{\imath\jmath}\frac{\varepsilon(N+8)}{4\sqrt{2}\pi\tau}+O(\tau^0)\,. \quad (4.53)$$

Removing the pole and integrating gives

$$\int_0^1 d\tau\left(\langle T_{ab}(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c+\mathcal{T}^{(2)}_{ab}\delta_{\imath\jmath}\frac{\varepsilon(N+8)}{4\sqrt{2}\pi\tau}\right)+\int_1^\infty d\tau\langle T_{ab}(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c$$
$$=\frac{N+8}{4\sqrt{2}\pi}\left(-\mathcal{T}^{(2)}_{ab}\delta_{\imath\jmath}\varepsilon\left(1-\log 2\right)+\mathcal{T}^{(3)}_{ab;\imath\jmath}\left(1+\varepsilon\left(\frac{N+6}{N+8}\log 2+\frac{N^2-5N-38}{2(N+8)^2}\right)\right)\right)\,. \quad (4.54)$$

This time the pole comes from the $\hat{\phi}$ contribution to the $\hat{t}_\imath\times\hat{t}_\jmath$ OPE,

$$\frac{\mathcal{N}^2_{\hat{t}}\lambda_{\hat{t}\hat{t}\hat{\phi}}b_{T\hat{\phi}}}{\tau^{2-\Delta_{\hat{\phi}}}} = -\frac{\varepsilon(N+8)}{4\sqrt{2}\pi\tau}+O(\varepsilon^2)\,. \quad (4.55)$$

We therefore integrate the left-hand side of the above expression. Combining the result with (4.54), we find

$$\int_{-\infty}^{\infty} d\tau\langle T_{ab}(0,1)\hat{t}_\imath(0)\hat{t}_\jmath(\tau)\rangle_c = \left(\mathcal{T}^{(3)}_{ab;\imath\jmath}-\mathcal{T}^{(2)}_{ab}\delta_{\imath\jmath}\right)\frac{N+8}{2\sqrt{2}\pi}\Bigg(1$$
$$+\varepsilon\left(\frac{N+6}{N+8}\log 2+\frac{N^2-5N-38}{2(N+8)^2}\right)\Bigg)\,. \quad (4.56)$$

The terms that come from the broken symmetry generator acting on the bulk operator T_{ab} are

$$\delta_{a\jmath}\langle T_{Nb}(0,1)\hat{t}_\imath(0)\rangle-\delta_{aN}\langle T_{\jmath b}(0,1)\hat{t}_\imath(0)\rangle+(a\leftrightarrow b)\,. \quad (4.57)$$

The end result of calculating these terms is an expression identical to the right-hand side of (4.56), and therefore this final example holds too.

4.3 $\int\langle\mathcal{O}\hat{\mathcal{O}}\hat{\mathcal{O}}\hat{t}\rangle \sim \langle\mathcal{O}\hat{\mathcal{O}}\hat{\mathcal{O}}\rangle$

A general bulk-defect-defect-defect correlator has the form

$$\langle\mathcal{O}_1(x_1)\hat{\mathcal{O}}_2(\tau_2)\hat{\mathcal{O}}_3(\tau_3)\hat{\mathcal{O}}_4(\tau_4)\rangle = \frac{F_{\mathcal{O}_1\hat{\mathcal{O}}_2\hat{\mathcal{O}}_3\hat{\mathcal{O}}_4}(u,v)}{|x_{1\perp}|^{\Delta_1}|\tau_{23}|^{\Delta_2+\Delta_3-\Delta_4}|\tau_{24}|^{\Delta_2-\Delta_3+\Delta_4}|\tau_{34}|^{-\Delta_2+\Delta_3+\Delta_4}}\,, \tag{4.58}$$

where the function $F_{\mathcal{O}_1\hat{\mathcal{O}}_2\hat{\mathcal{O}}_3\hat{\mathcal{O}}_4}$ now depends on the two cross-ratios

$$u = \frac{(\tau_{12}^2 + x_{1\perp}^2)\tau_{34}^2}{(\tau_{13}^2 + x_{1\perp}^2)\tau_{24}^2}\,, \quad v = \frac{(\tau_{14}^2 + x_{1\perp}^2)\tau_{23}^2}{(\tau_{13}^2 + x_{1\perp}^2)\tau_{24}^2}\,. \tag{4.59}$$

It is also useful to use a complex cross-ratio z that satisfies

$$z\bar{z} = u\,, \quad (1-z)(1-\bar{z}) = v\,. \tag{4.60}$$

We consider $\langle\phi_a\hat{t}_\imath\hat{t}_\jmath\hat{t}_k\rangle_c$, which takes the form

$$\langle\phi_a(0,x_{1\perp})\hat{t}_\imath(-1)\hat{t}_\jmath(1)\hat{t}_k(\tau)\rangle_c = -(\delta_{a\imath}\delta_{\jmath k} + \text{perms})\frac{\varepsilon\sqrt{N+8}}{8\pi^3(1+x_{1\perp}^2)(1-t)^2}z\bar{z}D(z,\bar{z})\,, \tag{4.61}$$

where Bloch-Wigner function is given by

$$D(z,\bar{z}) = \frac{1}{z-\bar{z}}\left(2\,\mathrm{Li}_2(z) - 2\,\mathrm{Li}_2(\bar{z}) + \log z\bar{z}\log\frac{1-z}{1-\bar{z}}\right)\,. \tag{4.62}$$

The function in (4.61) has no poles in τ at this order in ε, but rather one must take care of branch cuts when integrating. We proceed by noticing that the configuration used is even in τ, so we therefore restrict the integral to the positive real line. The correlator has branch cuts at $\tau = \pm 1$, so we write

$$\begin{aligned}\int_0^\infty d\tau\langle\phi_a(0,x_{1\perp})\hat{t}_\imath(-1)\hat{t}_\jmath(1)\hat{t}_k(\tau)\rangle_c = \int_0^1 d\tau\langle\phi_a(0,x_{1\perp})\hat{t}_\imath(-1)\hat{t}_\jmath(1)\hat{t}_k(\tau)\rangle_c \\ + \int_1^\infty\langle\phi_a(0,x_{1\perp})\hat{t}_\imath(-1)\hat{t}_\jmath(1)\hat{t}_k(\tau)\rangle_c\,.\end{aligned} \tag{4.63}$$

Calculating the primitive and evaluating it in four regimes, $\tau\to\infty$, $\tau\to 1^+$, $\tau\to 1^-$ and $\tau\to 0$, takes care of the branch cut at $\tau = 1$. After making use of the inversion and reflection identities for the di- and trilogarithm, one finds that

$$\begin{aligned}\int_{-\infty}^\infty d\tau\langle\phi_a(0,x_{1\perp})\hat{t}_\imath(-1)\hat{t}_\jmath(1)\hat{t}_k(\tau)\rangle_c = -(\delta_{a\imath}\delta_{\jmath k} + \text{perms})\frac{\varepsilon\sqrt{N+8}}{64\pi^2|x_{1\perp}|}\Bigg(\pi^2 \\ - 4\arccos^2\frac{2|x_{1\perp}|}{1+|x_{1\perp}|^2}\Bigg)\,.\end{aligned} \tag{4.64}$$

Even though (4.61) has no poles in τ at leading order in ε, the contact terms are $O(\varepsilon)$ and need to be included. The contact terms arise from the $\hat{\phi}$ contribution to the $\hat{t}_\imath(-1) \times \hat{t}_k(\tau)$ and $\hat{t}_\jmath(1) \times \hat{t}_k(\tau)$ OPE's. Therefore, we find the contact term contributions to be

$$2\frac{\mathcal{N}_{\hat{t}}^2 \lambda_{\hat{t}\hat{t}\hat{\phi}}}{\gamma_{\hat{\phi}}} \Big(\delta_{\imath k} \langle \phi_a(0, x_{1\perp}) \hat{\phi}(-1) \hat{t}_\jmath(1) \rangle_c + \delta_{\jmath k} \langle \phi_a(0, x_{1\perp}) \hat{t}_\imath(-1) \hat{\phi}(1) \rangle_c \Big) = \\ (\delta_{a\jmath}\delta_{\imath k} + \delta_{a\imath}\delta_{\jmath k}) \frac{\varepsilon\sqrt{N+8}}{64\pi^2 |x_{1\perp}|} \left(\pi^2 - 4 \arccos^2 \frac{2|x_{1\perp}|}{1+|x_{1\perp}|^2} \right) . \quad (4.65)$$

Finally, we need the terms from the broken symmetry generator acting on ϕ_a. We have

$$\delta_{ak} \langle \phi_N(0, x_{1\perp}) \hat{t}_\imath(-1) \hat{t}_\jmath(1) \rangle_c - \delta_{aN} \langle \phi_k(0, x_{1\perp}) \hat{t}_\imath(-1) \hat{t}_\jmath(1) \rangle_c = \\ \delta_{ak}\delta_{\imath\jmath} \frac{\varepsilon\sqrt{N+8}}{64\pi^2 |x_{1\perp}|} \left(\pi^2 - 4 \arccos^2 \frac{2|x_{1\perp}|}{1+|x_{1\perp}|^2} \right) . \quad (4.66)$$

Combining this with (4.64) and (4.65), we find that the expression vanishes as expected.

5 Conclusion

Through the breaking of an internal symmetry, the presence of a conformal defect introduces a distinguished set of operators with support on the defect. The tilt operators modify the current conservation equation and lead to powerful relations among connected correlation functions, as well as novel constraints on the CFT data. We develop a framework that allows us to derive relations among CFT data in complete generality by analyzing correlation functions that involve up to a single bulk insertion and up to three defect insertions.

We put our relations to use in two settings. First, the 1/2 BPS Maldacena-Wilson loop in $\mathcal{N} = 4$ SYM. In this setting, our constraints should hold at all values of the 't Hooft coupling and this is exactly what we find with the relation in (3.4). While other relations should hold for higher point functions, in practice it is more straightforward to work perturbatively at either weak- or strong-coupling. At weak-coupling, we demonstrate the power of our integral identities through self-consistency. With an ansatz of the forms of the three-point function of a 1/2 BPS operator in the bulk and two tilt operators on the defect, we are able to use some existing results to determine CFT data at leading order when the bulk operator has charge J, as well as some constraints at next-to-leading order.

After making use of our integral identities to highlight discrepancies among existing results at both weak- and strong-coupling, we hope that the tools we have developed will be useful in verifying the self-consistency of existing results, as well as being a valuable way of determining new CFT data in the future through their incorporation in the bootstrap program.

The second setting we study is the magnetic line defect in the $O(N)$ model in $d = 4 - \varepsilon$ dimensions. Here, we find that we are able to verify a wide array of CFT data that already exists in the literature. Moreover, new CFT data (also in [13]) that is calculated using traditional Feynman diagram techniques is found to be compatible with the constraints that we have derived.

Similar integral identities can also be constructed for displacement operators instead of tilts. These follow from the broken conformal symmetry generators and, as shown in [14], often lead to even more informative constraints than those derived from tilts. Related constructions have also been discussed in [13]. Another important aspect concerns anomalies. They play a crucial role in [14], and although we have not analyzed them in detail here, they are expected to be as important in the present bulk–defect setting as well. A further interesting direction arises from a geometric perspective. Tilts correspond to defect exactly marginal operators, allowing deformations along a defect conformal manifold. When the bulk theory also contains exactly marginal operators, one can integrate bulk insertions in the presence of the defect and study the richer structure of the combined bulk-defect conformal manifold.

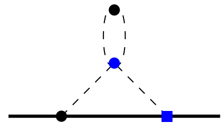
Acknowledgements

We are grateful to Nadav Drukker and Petr Kravchuk for their collaborative work in [14], from which some results are presented here and whose ideas and techniques we have adopted throughout this paper. This project would not have been possible without that work. We also thank them for many inspiring and helpful discussions. We are especially indebted to N. Drukker for reviewing the draft and providing valuable suggestions. Z.K. is supported by ERC-2021-CoG—BrokenSymmetries 101044226.

A Feynman Integrals and Correlation Functions

We compute Feynman integrals and two- and three-point functions, both with a single bulk operator and defect operators with dimensions close to one.

First, we consider the following diagram that contributes to the bulk-defect two-point function that involves a bulk operator of dimension close to two,



$$\propto \int \frac{d\tau_3 d^d x_4}{\left(\left(x_{14}^2\right)^2 x_{24}^2 x_{34}^2\right)^{\Delta_\phi}}\,, \tag{A.1}$$

where black circles denote external operators, and blue circles and squares denote bulk and defect interaction vertices respectively. We take $x_{1,4} = (\tau_{1,4}, x_{1,4\perp})$ and $x_{2,3} = (\tau_{2,3}, 0)$. Also,

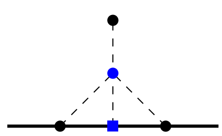
$\Delta_\phi = 1 - \frac{\varepsilon}{2}$ since these integrals are computed for bare operators. This integral is divergent in $d = 4$ and must therefore be computed in $d = 4 - \varepsilon$. The diagram is $O(\lambda)$, which means that it is enough to calculate the integral to $O(\varepsilon^0)$. To this end, we introduce Schwinger parameters u_1, u_2 and u_3. It is then trivial to carry out the integrals over τ_3 and x_4. One of the Schwinger parameters, say u_3, can be integrated by first introducing a resolution of the identity, $1 = \int_0^\infty dq\, \delta(q - u_3)$, and then rescaling the Schwinger parameters as $u_i \to q u_i$. The parameter q is easy to integrate away and the remaining delta function $\delta(1 - u_3)$ allows the u_3 integral to be done trivially. To progress, we need to identify where the integral is divergent using the procedure of analytic regularization as outlined in [38, 39]. The only divergence is when $u1 \to \infty$. The prescription to deal with this is to introduce another delta function, $1 = \int_0^\infty dq\, \delta(q - u_1^{-1})$, rescale $u_1 \to q^{-1} u_1$ and then partially integrate with respect to q. It is straightforward to check that the surface terms vanish and what remains is a factor proportional to $\frac{1}{\varepsilon}$ and a convergent integral that can be expanded i ε and evaluated term-by-term. The final result is

$$\int \frac{d\tau_3 d^d x_4}{\left(\left(x_{14}^2\right)^2 x_{24}^2 x_{34}^2\right)^{\Delta_\phi}} = \frac{\pi^3}{|x_{1\perp}|^{1-2\varepsilon} (x_{12}^2)^{1-\frac{\varepsilon}{2}}} \left(\frac{2}{\varepsilon} + 2 + 4 \log 2 - \gamma_E - \log \pi \right) . \tag{A.2}$$

This allows us to compute $b_{T\hat{t}}$ and $b_{T\hat{\phi}}$, which are given in (4.16) and (4.49) respectively.

Next, we calculate the connected diagram that contributes to the bulk-defect-defect three-point function when the bulk operator has dimension close to either one or two. We place the bulk operator at $x_0 = (\tau_0, x_{0\perp})$ and the defect operators at $x_{1,2} = (\tau_{1,2}, 0)$. In this setup, there is a single cross-ratio given by χ, defined in (2.50).

When the dimension of the bulk operator is close to one, the connected diagram that appears in this correlator is finite in $d = 4$ and is given by



$$\propto \int \frac{d\tau_3 d^4 x_4}{x_{04}^2 x_{14}^2 x_{24}^2 x_{34}^2} = \frac{\pi^3}{2} \frac{\mathcal{I}_{1,2}(\chi)}{|x_{0\perp}| \tau_{12}^2} , \tag{A.3}$$

where $x_3 = (\tau_3, 0)$ and $x_4 = (\tau_4, x_{4\perp})$ are the points being integrated over. We proceed to evaluate this integral by first taking $\tau_1 \to \infty$, and computing

$$\int \frac{d\tau_3 d^4 x_4}{x_{04}^2 x_{24}^2 x_{34}^2} = \pi^3 \frac{\pi^2 - 4 \arctan^2\left(\frac{\tau_{02}}{|x_{0\perp}|}\right)}{2|x_{0\perp}|} . \tag{A.4}$$

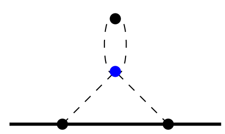
To evaluate this integral, one first introduces Schwinger parameters u_1, u_2, u_3. It is then straightforward to integrate over τ_3, τ_4 and $x_{4\perp}$. To integrate over the Schwinger parameters, it is useful to turn the integral into a projective integral by introducing $1 = \int_0^\infty dq\, \delta(q - u1 - u2 - u3)$ and rescaling each $u_i \to q u_i$. It is elementary to now integrate over the auxiliary variable q, while the delta function makes one of the Schwinger parameter integrals, say u_3, trivial. The final integrals over $0 < u_{1,2} < 1$ subject to $u_1 + u_2 < 1$ are less straightforward, but can be calculated in Mathematica. We are then able to reconstruct the original integral and conclude

$$\mathcal{I}_{1,2}(\chi) = \pi^2 - 4 \arccos^2\left(\sqrt{\chi}\right) . \tag{A.5}$$

When combining this result with the disconnected diagrams, which have been have been previously calculated in [37, 40], we find the following correlators,

$$\begin{aligned} F_{\phi_a \hat{t}_\imath \hat{t}_\jmath} &= -\delta_{a1}\delta_{\imath\jmath} \frac{(N+8)^{\frac{3}{2}}}{8\pi^2\chi} \left(1 + \varepsilon\left(\frac{7N^2+55N+190}{4(N+8)^2} + \frac{\log 2}{2} - \frac{\mathcal{I}_{1,2}(\chi)}{2(N+8)}\right)\right) , \\ F_{\phi_a \hat{\phi} \hat{t}_\imath} &= \delta_{a\imath} \frac{\mathcal{I}_{1,2}(\chi)}{8\pi\chi}\varepsilon , \\ F_{\phi_a \hat{\phi}\hat{\phi}} &= -\delta_{a1}\frac{\sqrt{N+8}}{2\chi}\left(1 + \varepsilon\left(\frac{N^2-3N-22}{4(N+8)^2} - \log\left(\frac{\chi}{\sqrt{2}}\right) - \frac{3\mathcal{I}_{1,2}(\chi)}{2(N+8)}\right)\right) . \end{aligned} \tag{A.6}$$

Finally, we consider the case where the dimension of the bulk operator is close to two. This configuration also has a new connected diagram that we need to calculate. It is given by



$$\propto \int \frac{d^d x_3}{\left(\left(x_{03}^2\right)^2 x_{13}^2 x_{23}^2\right)^{\Delta_\phi}} , \tag{A.7}$$

where $x_3 = (\tau_3, x_{3\perp})$ is the point being integrated over. This integral is also divergent in $d = 4$ and must be computed in $d = 4 - \varepsilon$. Since the diagram is $\mathcal{O}(\lambda)$, it suffices to compute this integral to $\mathcal{O}(\varepsilon^0)$. After we introduce the three Schwinger parameters $u_{1,2,3}$, it is straightforward to integrate τ_3 along the line and $x_{3\perp}$ over the $(d-1)$-dimensional bulk. To proceed, we introduce $1 = \int_0^\infty dq \delta(q - u3)$ and then rescale $u_i \to q u_i$. The integral over q is easy and the delta function allows us to trivially integrate over u_3. Once again, to progress we need to determine where the integral is divergent and make use of analytic regularization. As before, the only place where the integral is divergent is the contribution from $u_1 \to \infty$. We therefore deal with this by introducing another delta function, $1 = \int_0^\infty dq\, \delta\left(q - u_1^{-1}\right)$, rescaling $u_1 \to u_1 q^{-1}$ and partially integrating with respect to q. After determining that the surface terms vanish, we are able to expand the now convergent integral and evaluate term-by-term. The final result is

$$\int \frac{d^d x_4}{\left(\left(x_{14}^2\right)^2 x_{24}^2 x_{34}^2\right)^{\Delta_\phi}} = \frac{\pi^2}{x_{12}^{2-2\varepsilon} x_{13}^{2-2\varepsilon} t_{23}^{\varepsilon}} \left(\frac{2}{\varepsilon} + 2 - \gamma_E - \log\pi\right) . \tag{A.8}$$

We now report all non-zero correlation functions involving insertions of either s or T_{ab} in the bulk with two defect insertions with operators of dimension close to one,

$$\begin{aligned}F_{s\hat{\phi}\hat{\phi}} =&\sqrt{\frac{2}{N}}\Bigg(1+\frac{N+8}{8\chi}-\varepsilon\Bigg(\frac{7N+50}{2(N+8)}+\frac{13N+38}{16(N+8)\chi}-3\frac{1+4\chi}{4\chi}\log 2\\&+\left(\frac{N+8}{8\chi}-\frac{N+2}{2(N+8)}\right)\log\chi+\frac{3\mathcal{I}_{1,2}(\chi)}{8\chi}\Bigg)\Bigg),\end{aligned} \tag{A.9}$$

$$\begin{aligned}F_{S\hat{t}_\imath\hat{t}_\jmath} =&\delta_{\imath\jmath}\frac{N+8}{2\pi^2\sqrt{2N}}\Bigg(1+\frac{N+8}{8\chi}-\varepsilon\Bigg(\frac{13N+38}{2(N+8)^2}-\frac{3N^2+16N+68}{16(N+8)\chi}\\&-\left(1+\frac{3}{4\chi}\right)\log 2-\frac{N+2}{2(N+8)}\log\chi+\frac{\mathcal{I}_{1,2}(\chi)}{8\chi}\Bigg)\Bigg),\end{aligned} \tag{A.10}$$

$$\begin{aligned}F_{T_{ab}\hat{\phi}\hat{\phi}} =&\mathcal{T}^{(2)}_{ab}\sqrt{2}\Bigg(1+\frac{N+8}{8\chi}-\varepsilon\Bigg(\frac{3N+25}{N+8}-\frac{N^2-5N-38}{16(N+8)\chi}\\&-\left(3+\frac{N+6}{8\chi}\right)\log 2+\left(\frac{N+8}{8\chi}-\frac{1}{N+8}\right)\log\chi+\frac{3\mathcal{I}_{1,2}(\chi)}{8\chi}\Bigg)\Bigg),\end{aligned} \tag{A.11}$$

$$F_{T_{ab}\hat{t}_\imath\hat{\phi}} =\mathcal{T}^{(1)}_{ab;\imath}\frac{\sqrt{N+8}}{2\pi\sqrt{2}}\left(1-\varepsilon\left(\frac{5N^2+103N+438}{4(N+8)^2}-2\log 2-\frac{1}{N+8}\log\chi+\frac{\mathcal{I}_{1,2}(\chi)}{8\chi}\right)\right), \tag{A.12}$$

$$F_{T_{ab}\hat{t}_\imath\hat{t}_\jmath} =\mathcal{T}^{(3)}_{ab;\imath\jmath}\frac{N+8}{2\pi^2\sqrt{2}}\left(1+\varepsilon\left(\frac{N^2-5N-38}{2(N+8)^2}+\log 2+\frac{1}{N+8}\log\chi\right)\right)+ \tag{A.13}$$

$$\mathcal{T}^{(2)}_{ab}\delta_{\imath\jmath}\frac{(N+8)^2}{16\sqrt{2}\pi^2\chi}\left(1+\varepsilon\left(\frac{2N^2+12N+34}{(N+8)^2}+\frac{N+6}{N+8}\log 2-\frac{\mathcal{I}_{1,2}(\chi)}{N+8}\right)\right), \tag{A.14}$$

where the tensor structures appearing above are defined in (4.43), (4.50) and (4.52).

B Connected correlators

As stated in Section 1, the definition of the connected correlator is given in (1.6),

$$\begin{aligned}&\langle\mathcal{O}_{\alpha_1}(x_1)\cdots\mathcal{O}_{\alpha_m}(x_m)\hat{t}_{i_1}(\tau_1)\cdots\hat{t}_{i_n}(\tau_n)\rangle_c\\=&\frac{\delta}{\delta r^{\alpha_1}(x_1)}\cdots\frac{\delta}{\delta r^{\alpha_m}(x_m)}\frac{\delta}{\delta w^{i_1}(\tau_1)}\cdots\frac{\delta}{\delta w^{i_n}(\tau_n)}\log Z[r,w]\Big|_{r=w=0}.\end{aligned} \tag{B.1}$$

Consider the set of operators

$$S=\{\mathcal{O}_{\alpha_1}(x_1),\cdots,\mathcal{O}_{\alpha_m}(x_m),\hat{\mathcal{O}}_{i_1}(\tau_1),\cdots,\hat{\mathcal{O}}_{i_n}(\tau_n)\} \tag{B.2}$$

which collects all bulk and defect operators appearing in the correlators, with tilts generalized to defect operators $\hat{\mathcal{O}}$. With this notation, (B.3) may be rewritten as

$$\langle\mathcal{O}_{\alpha_1}(x_1)\cdots\mathcal{O}_{\alpha_m}(x_m)\hat{\mathcal{O}}_{i_1}(\tau_1)\cdots\hat{\mathcal{O}}_{i_n}(\tau_n)\rangle_c=\sum_k\sum_{\substack{S_1\cup\cdots\cup S_k=S\\S_i\cap S_j=\varnothing}}(-1)^{k-1}(k-1)!\langle S_1\rangle\cdots\langle S_k\rangle\,, \tag{B.3}$$

where S_i denotes a proper subset of S. We now analyze several simple cases as illustrative examples.

m=0

When there are no bulk operator insertions, we consider defect two-, three- and four-point functions. Since defect one-point functions vanish, the case of the two-point function is trivial,

$$\langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle_c = \langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle - \langle \hat{\mathcal{O}}_{i_1}(\tau_1)\rangle\langle \hat{\mathcal{O}}_{i_2}(\tau_2)\rangle = \langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle \,. \tag{B.4}$$

We see that the full defect two-point function is connected. For the same reason, the full three-point function is also connected,

$$\langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle_c = \langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle \,. \tag{B.5}$$

As for the four-point function, the disconnected components are the three products of two-point functions that can be constructed from pairs of operators that are being considered,

$$\begin{aligned}
&\langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\hat{\mathcal{O}}_{i_4}(\tau_4)\rangle_c \\
=&\langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\hat{\mathcal{O}}_{i_4}(\tau_4)\rangle - \langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle\langle \hat{\mathcal{O}}_{i_3}(\tau_3)\hat{\mathcal{O}}_{i_4}(\tau_4)\rangle \\
&- \langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle\langle \hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_4}(\tau_4)\rangle - \langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_4}(\tau_4)\rangle\langle \hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle \,.
\end{aligned} \tag{B.6}$$

It can be that some of these two-point functions are zero due to symmetry requirements, but there will be at least one pairing that is non-vanishing since otherwise the whole four-point function would be zero.

m=1

In the presence of the defect, bulk operators admit a non-zero one-point function which is naturally connected,

$$\langle \mathcal{O}_{I_1}(x_1)\rangle_c = \langle \mathcal{O}_{I_1}(x_1)\rangle \,. \tag{B.7}$$

Once again, thanks to vanishing defect one-point functions, the bulk-defect two-point function is connected,

$$\langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_1}(\tau_1)\rangle_c = \langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_1}(\tau_1)\rangle \,. \tag{B.8}$$

For the case with two defect insertions, the only non-zero disconnected contribution can come from the product of the bulk one-point function with the defect two-point function,

$$\langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle_c = \langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle - \langle \mathcal{O}_{I_1}(x_1)\rangle\langle \hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle \,. \tag{B.9}$$

Finally, when we have three operators on the defect there are two types of disconnected contributions. The first is the product of the bulk one-point function with the defect three-point function, and the other is the product of a bulk-defect two-point function with a defect two-point function. Since there are three defect operators, the latter contributes three terms,

$$\begin{aligned} \langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle_c &= \langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle \\ -\langle \mathcal{O}_{I_1}(x_1)\rangle\langle\hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle &- \langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_1}(\tau_1)\rangle\langle\hat{\mathcal{O}}_{i_2}(\tau_2)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle \\ -\langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle\langle\hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle &- \langle \mathcal{O}_{I_1}(x_1)\hat{\mathcal{O}}_{i_3}(\tau_3)\rangle\langle\hat{\mathcal{O}}_{i_1}(\tau_1)\hat{\mathcal{O}}_{i_2}(\tau_2)\rangle\,. \end{aligned} \tag{B.10}$$

References

[1] J. L. Cardy, "Boundary Conditions, Fusion Rules and the Verlinde Formula," *Nucl. Phys. B* **324** (1989) 581–596.

[2] D. McAvity and H. Osborn, "Conformal field theories near a boundary in general dimensions," *Nucl. Phys. B* **455** (1995) 522–576, `cond-mat/9505127`.

[3] M. Billò, V. Gonçalves, E. Lauria, and M. Meineri, "Defects in conformal field theory," *JHEP* **04** (2016) 091, `arXiv:1601.02883`.

[4] R. G. Leigh and M. J. Strassler, "Exactly marginal operators and duality in four-dimensional $\mathcal{N} = 1$ supersymmetric gauge theory," *Nucl. Phys. B* **447** (1995) 95–136, `hep-th/9503121`.

[5] D. Green, Z. Komargodski, N. Seiberg, Y. Tachikawa, and B. Wecht, "Exactly Marginal Deformations and Global Symmetries," *JHEP* **06** (2010) 106, `arXiv:1005.3546`.

[6] J. Gomis, P.-S. Hsin, Z. Komargodski, A. Schwimmer, N. Seiberg, and S. Theisen, "Anomalies, Conformal Manifolds, and Spheres," *JHEP* **03** (2016) 022, `arXiv:1509.08511`.

[7] A. Giambrone, A. Guarino, E. Malek, H. Samtleben, C. Sterckx, and M. Trigiante, "Holographic evidence for nonsupersymmetric conformal manifolds," *Phys. Rev. D* **105** no. 6, (2022) 066018, `arXiv:2112.11966`.

[8] N. Drukker, Z. Kong, and G. Sakkas, "Broken global symmetries and defect conformal manifolds," *Phys. Rev. Lett.* **129** no. 20, (2022) 201603, `arXiv:2203.17157`.

[9] C. P. Herzog and V. Schaub, "The Tilting Space of Boundary Conformal Field Theories," `arXiv:2301.10789`.

[10] G. Sakkas, "Inversion and Integral Identities in dCFTs," `arXiv:2403.05243`.

[11] J. Belton, N. Drukker, Z. Kong, and A. Stergiou, "Fine spectrum from crude analytic bootstrap," *J. Phys. A* **58** no. 34, (2025) 345401, `arXiv:2503.07710`.

[12] V. Bashmakov, M. Bertolini, and H. Raj, "On non-supersymmetric conformal manifolds: field theory and holography," *JHEP* **11** (2017) 167, `arXiv:1709.01749`.

[13] B. Girault, M. F. Paulos, and P. van Vliet, "Consequences of symmetry-breaking on conformal defect data," arXiv:2509.26561.

[14] N. Drukker, Z. Kong, and P. Kravchuk *to appear* .

[15] N. Drukker and Z. Kong, "1/3 BPS loops and defect CFTs in ABJM theory," *JHEP* **06** (2023) 137, arXiv:2212.03886.

[16] G. Bliard, D. H. Correa, M. Lagares, and I. Salazar Landea, "Bootstrapping line defects in AdS_3/CFT_2," *JHEP* **01** (2025) 131, arXiv:2410.02685.

[17] Z. Kong, "Integral Identities from Symmetry Breaking of Conformal Defects," in *14th International Workshop on Lie Theory and Its Applications in Physics*. 9, 2025. arXiv:2509.23797.

[18] D. Friedan and A. Konechny, "Curvature formula for the space of 2-d conformal field theories," *JHEP* **09** (2012) 113, arXiv:1206.1749.

[19] J. Padayasi, A. Krishnan, M. A. Metlitski, I. A. Gruzberg, and M. Meineri, "The extraordinary boundary transition in the 3d O(N) model via conformal bootstrap," *SciPost Phys.* **12** no. 6, (2022) 190, arXiv:2111.03071.

[20] A. Bzowski, P. McFadden, and K. Skenderis, "Scalar 3-point functions in CFT: renormalisation, beta functions and anomalies," *JHEP* **03** (2016) 066, arXiv:1510.08442.

[21] A. B. Zamolodchikov, "Three-point function in the minimal Liouville gravity," *Theor. Math. Phys.* **142** (2005) 183–196, arXiv:hep-th/0505063.

[22] D. Harlow, J. Maltz, and E. Witten, "Analytic Continuation of Liouville Theory," *JHEP* **12** (2011) 071, arXiv:1108.4417.

[23] J. M. Maldacena, "Wilson loops in large N field theories," *Phys. Rev. Lett.* **80** (1998) 4859–4862, hep-th/9803002.

[24] N. Drukker, D. J. Gross, and H. Ooguri, "Wilson loops and minimal surfaces," *Phys. Rev.* **D60** (1999) 125006, hep-th/9904191.

[25] L. F. Alday and J. Maldacena, "Comments on gluon scattering amplitudes via AdS/CFT," *JHEP* **11** (2007) 068, arXiv:0710.1060.

[26] J. Polchinski and J. Sully, "Wilson Loop renormalization group flows," *JHEP* **10** (2011) 059, arXiv:1104.5077.

[27] M. Beccaria, S. Giombi, and A. Tseytlin, "Non-supersymmetric Wilson loop in $\mathcal{N} = 4$ SYM and defect 1d CFT," arXiv:1712.06874.

[28] R. Brüser, S. Caron-Huot, and J. M. Henn, "Subleading Regge limit from a soft anomalous dimension," *JHEP* **04** (2018) 047, arXiv:1802.02524.

[29] G. Cuomo, Z. Komargodski, and A. Raviv-Moshe, "Renormalization Group Flows on Line Defects," *Phys. Rev. Lett.* **128** no. 2, (2022) 021603, arXiv:2108.01117.

[30] S. Giombi, R. Roiban, and A. A. Tseytlin, "Half-BPS Wilson loop and AdS_2/CFT_1," *Nucl. Phys.* **B922** (2017) 499–527, arXiv:1706.00756.

[31] S. Giombi and S. Komatsu, "More exact results in the Wilson loop defect CFT: bulk-defect OPE, nonplanar corrections and quantum spectral curve," arXiv:1811.02369.

[32] D. Artico, J. Barrat, and Y. Xu, "Perturbative bootstrap of the Wilson-line defect CFT: Bulk-defect-defect correlators," arXiv:2410.08273.

[33] S. Giombi and S. Komatsu, "Exact correlators on the Wilson loop in $\mathcal{N} = 4$ SYM: Localization, defect CFT, and integrability," *JHEP* **05** (2018) 109, arXiv:1802.05201.

[34] C. Beem, M. Lemos, P. Liendo, W. Peelaers, L. Rastelli, and B. C. van Rees, "Infinite chiral symmetry in four dimensions," *Commun. Math. Phys.* **336** no. 3, (2015) 1359–1433, arXiv:1312.5344.

[35] N. Drukker, S. Giombi, R. Ricci, and D. Trancanelli, "Wilson loops: From four-dimensional SYM to two-dimensional YM," *Phys. Rev. D* **77** (2008) 047901, arXiv:0707.2699.

[36] M. Beccaria and A. A. Tseytlin, "On the structure of non-planar strong coupling corrections to correlators of BPS Wilson loops and chiral primary operators," *JHEP* **01** (2021) 149, arXiv:2011.02885.

[37] A. Gimenez-Grau, "Probing magnetic line defects with two-point functions," arXiv:2212.02520.

[38] E. Panzer, "On hyperlogarithms and Feynman integrals with divergences and many scales," *JHEP* **03** (2014) 071, arXiv:1401.4361.

[39] E. Panzer, *Feynman integrals and hyperlogarithms*. PhD thesis, Humboldt U., 2015. arXiv:1506.07243.

[40] A. Gimenez-Grau, E. Lauria, P. Liendo, and P. van Vliet, "Bootstrapping line defects with $O(2)$ global symmetry," *JHEP* **11** (2022) 018, arXiv:2208.11715.